

International Spillovers of U.S. Financial Volatility

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Abstract

We study the international spillover effects of U.S. bond and stock market volatility using a panel data set of seventeen developed countries. We find significant spillover impacts of U.S. financial market volatility on international output growth. Volatility of U.S. bonds with longer maturities is shown to have a more significant impact on international output growth than volatility of U.S. bonds with shorter maturities. Regardless of the maturity of the bonds, U.S. financial market volatility is shown to play a larger role statistically in explaining international output growth than a country's own financial market volatility.

Keywords: bond volatility, stock volatility, international output growth, spillovers

JEL: E4, F3, F4, G15

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1 Introduction

In this paper, we study the spillover effects of U.S. financial market volatility on international output growth. Within a country, when financial uncertainty is high, portfolios generally favor safe assets, precautionary saving is high, real investment is low, and this leads to low output growth. What is not clear however, is whether financial volatility within the U.S. can exhibit real effects internationally. We address this issue by employing dynamic panel regressions and panel vector autoregressions (VARs) to study whether realized volatility in the U.S. bond and stock markets impacts international industrial production growth using a panel data set of seventeen developed countries.¹

This paper's key result is that changes in U.S. financial market volatility have significant spillover effects on international industrial production growth.² Moreover, we also find that volatility of U.S. bonds with longer maturities tends to have a more significant impact on international output growth than volatility of U.S. bonds with shorter maturities. Regardless of the maturity of the bonds, U.S. financial market volatility is more important statistically in explaining movements in international industrial production growth than financial market volatility within one's own country. These results are robust to various controls, sub-samples, and model specifications.

Our consideration of stock market volatility follows in part, from the existing literature that shows during times of high stock market volatility, output growth is low on a country-by-country basis (e.g., Schwert (1989), Schwert (1990), Engle et al. (2013), Asgharian et al. (2013), Conrad and Loch (2015), Ludvigson et al. (2016)), as well as in a panel setting without spillovers (e.g., Vu (2015)). Our emphasis on studying the spillover effects is partly motivated by a large literature that documents the co-movement of business cycle variables across countries (e.g., Kose et al. (2008), Crucini et al. (2011), Apostolou and Beirne (2017)). We consider bond market volatility, in addition to stock market volatility, given the connection between bond and stock markets, as

¹We use industrial production growth and output growth interchangeably. The data used in our analysis is industrial production growth.

²In the panel analyses, the term *international* refers to all countries other than the U.S. in our sample.

documented in the literature (e.g., Connolly et al. (2005)), as well as the documented relationship between U.S. bond market volatility and U.S. output growth (e.g., Cremers et al. (2017)). Our paper complements this literature by being the first, to the best of our knowledge, to document the spillover effects of both U.S. bond and stock market volatility on international output growth.

We depart from the finance literature, however, by focusing on realized financial market volatility rather than implied financial market volatility because implied volatility is not widely available across countries. We construct the financial market volatility variables by first converting daily government bond and stock market indices into daily bond and stock market returns for a set of seventeen developed countries. We then calculate the standard deviation of the daily bond and stock market return data within each month to construct realized monthly bond and stock market volatility for each country in our sample, which spans from 1998M3 to 2017M2.³

Our paper also belongs to a growing literature that studies the interplay of volatility across multiple financial markets (e.g., Hamao et al. (1990), Lin and Tamvakis (2001), Jones and Wilson (2004), and Diebold and Yilmaz (2009)). We add to this literature by documenting significant spillover effects of volatility on real economic activity of multiple economies. Lastly, our paper contributes to a literature that studies the relationship between U.S. macroeconomic news and U.S. bond market volatility (e.g., Jones et al. (1998)) and complements a large and growing theoretical literature that studies the extent to which risk and uncertainty can have real impacts on output growth (e.g., Fernandez-Villaverde et al. (2011)).

The remainder of the paper is organized as follows. Section 2 describes the data sources and the construction of the data used in our empirical analysis. Section 3 outlines the dynamic panel methodology and summarizes the main empirical results. Section 4 discusses the impulse responses and variance decomposition results from a panel VAR framework. Section 5 concludes.

³We construct monthly financial market volatility to be consistent with the frequency of industrial production. Further details on the data and construction are outlined in Section 2.

2 The Data

The data used in our empirical analysis are monthly and have a maximal span from 1998M3 to 2017M2, but the raw data are either daily or monthly. Our sample is composed of seventeen developed countries and includes Austria, Belgium, Canada, Denmark, Finland, France, Germany, Ireland, Italy, Japan, the Netherlands, Norway, Portugal, Spain, Sweden, the United Kingdom, and the United States.⁴ Our raw data consist of industrial production, interest rates, economic policy uncertainty indices, composite leading indicators, an index of consumer sentiment, oil prices, VIX, stock market indices, and bond market indices. We obtain non-seasonally adjusted data when available and remove seasonality using our own methodology by taking a moving average of the current and twelve previous months of the variable under consideration.⁵

We obtain monthly industrial production (IP) data from the International Monetary Fund’s International Financial Statistics (IFS) database. The 3-month interest rates are taken from Datastream.⁶ Monthly economic policy uncertainty indices are taken from Baker et al. (2016).⁷ The monthly composite leading indicators (CLI) are taken from the Organisation for Economic Co-operation and Development (OECD) database. The University of Michigan’s Consumer Sentiment Index (MCI), the (daily) West Texas Intermediate (WTI) oil prices, and the Chicago Board Options Exchange (CBOE) Volatility Index (VIX) are taken from the FRED database at the Federal Reserve Bank of St. Louis. Daily stock market index data are from the Morgan Stanley Capital International (MSCI) index and are denominated in U.S. dollars (USD). Daily government bond index data are from the Bank of America Merrill Lynch Bond Index database and are also denominated in USD.⁸ The government bond indices are available at different

⁴The selection of countries was based on data availability.

⁵We avoid using the Census seasonal adjustment procedure because it impounds future information into today’s seasonally adjusted observations.

⁶The details for the interest rates series for all countries in our sample can be found in Appendix A.

⁷These indices can be downloaded at www.policyuncertainty.com. Economic policy uncertainty data for Japan are from Arbatli et al. (2017), following the approach of Baker et al. (2016). When economic policy uncertainty data are not available for countries, we use their corresponding regional series (e.g., EPU series for Europe).

⁸We provide additional details on the construction of these stock market and bond market indices in Appendix B.

maturities and include bonds with maturities of one to three years, three to five years, five to seven years, seven to ten years, and one or more years. Going forward, we will refer to the one-three year maturity as *three*, the three-five year maturity as *five*, the five-seven year maturity as *seven*, the seven-ten year maturity as *ten*, and the one or more years as *one plus*.

For our empirical analysis, we construct industrial production growth by taking the log first difference of the industrial production series. The change in the interest rate, the change in CLI, and the change in MCI are the first difference of each series. To construct monthly oil volatility, we calculate the standard deviation of the daily de-trended WTI (in log) within each month. To construct our measures of realized monthly bond and stock market volatility, we first normalize the data so that January 1, 2010 equals 100 and then convert the normalized series into daily bond and stock market returns.⁹ Within each month, we then calculate the standard deviation of the daily bond and stock market return data to construct the monthly realized bond and stock market volatility variables used in our analysis.

[Table 1 Here]

Table 1 presents the mean and standard deviation of realized bond and stock market volatility by country for the time period 1998M3 to 2017M2. Overall, the mean and standard deviation of stock market volatility are generally higher than the mean and standard deviation of bond market volatility. For some countries, the descriptive statistics for bond market volatility appear to be roughly similar across all maturities. Figure 1 plots realized bond (one plus) and stock market volatility over time for all countries in our sample, with G7 countries on the left and non-G7 countries on the right. Across all countries, stock markets appear to have been relatively more volatile than the corresponding bond markets, as also indicated in Table 1.

[Figure 1 Here]

We construct realized financial market volatility rather than implied financial market volatility, as is common in the finance literature because implied volatility is not widely available across

⁹We do this normalization to be consistent with the industrial production index.

countries. We hypothesize though, that our constructed measures of realized financial market volatility are still able to capture some of the variation generated by their implied counterparts. Figure 2 plots our measure of stock market volatility for the U.S. against the Chicago Board Options Exchange (CBOE) Volatility Index (VIX), one of the most commonly used measures of implied volatility in the literature.¹⁰ The correlation between these two series is 0.89, which suggests that our measure of realized stock market volatility is able to capture some of the information conveyed in implied volatility.¹¹ We also plot the VIX against our constructed measure of U.S. bond market volatility for bonds with a maturity of one or more years (one plus). The moderately low, albeit positive, correlation between these two series suggests that bond market volatility might contain information that is beyond what stock market volatility can convey, an issue that we will investigate in greater detail in the next sections.

[Figure 2 Here]

3 Dynamic Panel

This section addresses the central issue of the paper. Does U.S. financial market volatility exhibit spillover effects on international output growth? Our investigation of the impacts of U.S. financial market volatility includes both U.S. stock market volatility and U.S. bond market volatility. We first outline our baseline empirical methodology and then discuss our dynamic panel regression results.

3.1 Dynamic Panel Methodology

To study the impacts of U.S. financial market volatility on international industrial production growth, we follow Judson and Owen (1999) and use a least square dummy variable (LSDV)

¹⁰Data for the Chicago Board Options Exchange (CBOE) Volatility Index (VIX) are from the series VIXCLS from <https://fred.stlouisfed.org/series/VIXCLS> and was retrieved on March 7, 2018.

¹¹ Since corresponding VIX measures that are constructed in a methodologically consistent manner are not available for all countries in our sample, we use realized U.S. stock market volatility rather than the VIX in our empirical analysis to be consistent with the realized international stock market volatility variables. We also use realized bond market volatility rather than implied bond market volatility for all countries in our sample for similar reasons.

regression with heteroskedasticity and autocorrelation consistent (HAC) standard errors to estimate the following dynamic panel regression,¹²

$$\Delta \ln(IP_{i,t}) = \alpha_i + \sum_{j=1}^2 \delta_j \text{US Bond Vol}_{t-j} + \sum_{j=1}^2 \beta_j \text{US Stock Vol}_{t-j} + \sum_{j=1}^2 \phi_j V_{i,t-j} + \gamma X_t + \zeta Z_{i,t} + \varepsilon_{i,t} \quad (1)$$

where $\Delta \ln(IP_{i,t})$ is industrial production growth for country i in time t , *US Bond Vol* _{$t-j$} is the lag of U.S. bond market volatility, *US Stock Vol* _{$t-j$} is the lag of U.S. stock market volatility, $V_{i,t-j}$ is a set of lags of country i 's specific controls that includes industrial production growth, stock market volatility, bond market volatility, and the change in the interest rate, X_t is a set of U.S. specific controls that includes the economic policy uncertainty (U.S. EPU) index from Baker et al. (2016), the change in Michigan's Consumer Sentiment Index (Δ MCI), oil volatility, and industrial production growth, and $Z_{i,t}$ is a set of country i specific controls that includes the change in the composite leading indicator (Δ CLI) and the economic policy uncertainty (EPU) index from Baker et al. (2016). When the economic policy uncertainty data are not available for a specific country, we use EPU data for the region (e.g., Europe). Our choice of using a least square dummy variable (LSDV) regression instead of the Arellano and Bond (1991) estimator can be attributed to the fact that our panel is dominated by the time-series dimension (with a relatively large T). We also include country fixed effects to control for heterogeneity arising from a country's individual characteristics.

Our panel data set is balanced with a time span from 1998M3 to 2017M2. We exclude the U.S. as a country in our panel analysis because we are interested in the spillover effects of U.S. financial market volatility on international output growth. In this paper, the terms *country i* and *international* refer to all countries other than the U.S. in our sample. We select a lag length of two for each country in the dynamic panel regression because across all countries, two is the most common optimal lag length according to the Schwarz criterion (SBIC).¹³

¹²We consider various combinations of the control variables where our benchmark specification excludes the control variables in X_t and $Z_{i,t}$. We also adjust for small sample bias following Kiviet (1995).

¹³Using the benchmark specification variables for each country in our sample, we test for the optimal lag length using the SBIC with a maximal lag length of twelve months. Our most common optimal lag length across all countries in our sample is two. None of the international countries have an optimal lag length greater than

As a preamble to the dynamic panel analysis, we test for unit roots in our data by following Im et al. (2003) (IPS hereafter), who extend the standard Dickey-Fuller test to allow for cross-sectional heterogeneity in panel data, and find that we can reject the null hypothesis that all cross-sections contain unit roots. We provide the statistics for the IPS test in Appendix C.

3.2 The Impacts of Financial Market Volatility on Output Growth

In Table 2, we present the dynamic panel results where international industrial production growth is the dependent variable. We omit the term *international* in the table to save on space. We distinguish between the *international* and *U.S.* variables by explicitly identifying the *U.S.* variables in the table.¹⁴ In specification (1), the regressors only include the lags of U.S. bond market volatility and the lags of international industrial production growth as the controls. In specification (2), the regressors include the lags of U.S. stock market volatility along with the lags of international industrial production growth as the controls. In specifications (3) to (6), the regressors include the lags of U.S. bond market volatility, the lags of U.S. stock market volatility, and the lags of international industrial production growth (specification (3)), as well as the lags of international bond market volatility (specification (4)), the lags of international stock market volatility (specification (5)), and the lags of the change in the international interest rate (specification (6)). Across all specifications, U.S. and international bond market volatility are constructed using bonds with a maturity of one or more years (one plus). We consider specification (6) our benchmark specification.

[Table 2 Here]

Our key insight from Table 2 is that U.S. financial market volatility exhibits spillover effects on international output growth. More specifically, we find the lags of U.S. bond market volatility to be negatively and significantly associated with international industrial production growth.¹⁵ Moreover, this negative relationship is statistically significant even after controlling

two according to the SBIC.

¹⁴In all future tables, we do the same.

¹⁵Our result complements Cremers et al. (2017), who document a negative relationship between U.S. Treasury yield implied volatility and U.S. output growth.

for the lags of international industrial production growth, for the change in the international interest rate, and for volatility arising from the international bond markets, the international stock markets, and the U.S. stock market. While the level of significance is less pronounced on the first lag of U.S. bond market volatility, the addition of these controls does not alter the level of statistical significance of the coefficients on the second lag of U.S. bond market volatility. The result that bond market volatility (which is constructed from government bonds) is negatively associated with output growth is supported by the theoretical literature on the negative impact of fiscal volatility on economic growth. For example, Fernandez-Villaverde et al. (2011) show in a dynamic framework that an unexpected shock to fiscal volatility can negatively impact output growth.

We also find a statistically significant relationship between U.S. stock market volatility and international output growth in Table 2. In particular, the coefficient on the first lag of U.S. stock market volatility is positive and significant (except in specification (2)) whereas the coefficient on the second lag of U.S. stock market volatility is negative and significant (except in specification (6)).¹⁶ This result complements the literature that studies the impact of stock market volatility on output growth on a country-by-country basis. For example, Schwert (1989) and Schwert (1990) document a negative relationship between U.S. stock market volatility and U.S. output growth. Another example is Vu (2015), who documents a statistically significant negative relationship between stock market volatility and the country's own industrial production growth in a panel environment. In our dynamic panel analysis, however, we find this statistically significant relationship to mostly disappear once we include U.S. financial market volatility variables. Overall, the results in Table 2 imply that understanding U.S. financial market conditions can help us further understand the dynamics of international economies beyond what can be explained by a country's own domestic financial market conditions.

In Table 3, we introduce a number of additional controls to our benchmark dynamic panel

¹⁶We note that the coefficient on the first lag of U.S. stock volatility generally becomes statistically insignificant when additional controls are included in the regression. As a result, for the remainder of the paper, we do not focus our discussion on the first lag of U.S. stock volatility. We focus on the second lag of U.S. stock volatility in the paper instead, because when we consider volatility of bonds with shorter maturities rather than longer as in one plus, we find the second lag of U.S. stock volatility to be statistically significant when we include all controls.

regression. Specifically, we include a measure of economic policy uncertainty (EPU and U.S. EPU) in the spirit of Baker et al. (2016) to control for the fact that uncertainty arising from changes in economic policy can impact a firm’s investment decision and therefore can significantly impact future output growth (specifications (1) and (2)). To ensure that our measures of U.S. financial market uncertainty are not simply reflective of other sources of uncertainty in the economy, we also control for uncertainty arising from fluctuations of economic activity around its long term potential level (Δ CLI, specification (3)), from fluctuations in consumer confidence using the change in the Michigan Consumer Sentiment Index (Δ MCI, specification (4)), and from fluctuations in the commodity market using a measure of realized oil volatility (specification (5)). We also include U.S. industrial production growth (specification (6)) to control for the co-movement of business cycle variables across countries. Even with these additional country-specific and U.S.-specific controls, we still find a negative and significant spillover effect of U.S. bond market volatility (second lag) on international industrial production growth, and this impact is more pronounced than the international counterpart, as evidenced by the higher level of statistical significance.

[Table 3 Here]

3.3 Bond Market Volatility and the Term Structure

In this section, we study the extent to which different bond maturities can help explain movements in international output growth. To address this issue, we re-run the dynamic panel regression in Equation (1) under the benchmark specification, but instead of using U.S. and international bond market volatility that are constructed from bonds with maturities of one or more years (one plus), we consider bond market volatility constructed from bonds with maturities of one to three years (three), three to five years (five), five to seven years (seven), and seven to ten years (ten). Table 4 reports the dynamic panel regression results. Figure 3 plots the coefficients on the second lags of the various financial market variables for each specification in Table 4, along with their corresponding 95% confidence bands.

[Table 4 Here]

In Figure 3, U.S. bond market volatility appears to exhibit a negative and more statistically significant impact on international industrial production growth when it is constructed from bonds with longer maturities than shorter maturities. On the other hand, the second lag of U.S. stock market volatility appears to have a negative and significant impact on international output growth when we include U.S. and international bond market volatility constructed from shorter bond maturities. In stark contrast, none of the coefficients on the second lags of international bond and stock market volatility are significant at the 5% level. These contrasting patterns dovetail with our earlier result that statistically, U.S. financial market volatility, as a whole, is able to better explain movements in international industrial production growth than its international counterparts. We also include additional controls in the spirit of Table 3, but with bond market volatility constructed from bonds with maturities of three, five, seven, and ten years. To economize on space, we report these results in Appendix D.

[Figure 3 Here]

In sum, U.S. financial volatility has significant spillover effects on international output growth. In particular, volatility of U.S. bonds with longer maturities has a more statistically significant impact on international industrial production growth than volatility of U.S. bonds with shorter maturities. Regardless of the maturity of the bonds considered, the results suggest that U.S. financial market volatility is more important statistically than a country's own financial volatility in explaining movements in international output growth.

3.4 Panel Sub-sample Analysis

We consider a number of sub-sample analyses. First, we restrict our panel sample to the time period prior to the Great Recession (1998M3-2007M11). The panel results using this sub-sample are reported in Table 5 and correspond to our benchmark specification with different bond maturities (three, five, seven, ten, and one plus years) as in Table 4. Here, the coefficients

on U.S. stock volatility are positive, which indicate that volatility in the U.S. stock market generates positive international externalities during a mainly expansionary time period in the United States. Intuitively, high volatility during this boom period can be construed as a sign of market rebalancing toward more productive industries. Consequently, periods of ‘good’ U.S. stock volatility can have positive spillover impacts across countries. This is in stark contrast with the ‘bad’ U.S. stock volatility associated with heightened uncertainty during the Great Recession. The pre-Great Recession results with respect to U.S. bond market volatility are qualitatively consistent with the full sample results. Given this, our results in the previous sections do not appear to be driven solely by the Great Recession.

[Table 5 Here]

We next divide our sample into G7 countries and non-G7 countries to examine if our results are being driven by these subsets of countries. The G7 countries are Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States. Since U.S. bond volatility is a variable of interest in our analysis, we exclude the United States from the G7 sample. These results are reported in Table 6 and correspond to our benchmark specification with different bond maturities (three, five, seven, ten, and one plus years). The U.S. bond volatility results for the G7 and non-G7 countries are roughly consistent with our full sample U.S. bond volatility results. The results for the non-G7 countries are weaker, but the coefficient on the second lag of U.S. bond volatility remains negative. The opposite is the case for the second lag of U.S. stock volatility. The non-G7 country results are roughly consistent with our full sample U.S. stock volatility results and the results for the G7 countries are weaker, but the coefficient on the second lag of U.S. stock volatility remains negative. Although the G7 and non-G7 results appear to be slightly weaker statistically than our full sample results in some cases, they are qualitatively consistent. For example, the second lags of U.S. bond and stock market volatility are consistently negatively associated with international industrial production growth.

[Table 6 Here]

4 Panel VAR

In this section, we study the responses of international industrial production growth to U.S. and international financial market volatility shocks at different horizons using a panel vector autoregression (VAR) framework that is estimated by generalized method of moments (GMM). We first outline our panel VAR methodology and then we discuss the results from an impulse response analysis and a variance decomposition analysis. We find that the panel VAR results are generally consistent with the dynamic panel regression results in Section 3. Specifically, we find significant spillover effects of U.S. financial market volatility on international output growth over time. We also highlight the complementary roles of U.S. stock market volatility and U.S. bond market volatility at different maturities. In particular, the volatility of U.S. bonds with longer maturities appear to be better positioned statistically to explain movements in international industrial production growth than volatility of U.S. bonds with shorter maturities.

4.1 Panel VAR Methodology

We investigate how international industrial production growth responds to sudden one-standard-deviation increases in the innovations of U.S. and international financial market volatility at different horizons in a panel VAR framework. In the spirit of Love and Zicchino (2006), we estimate the following panel VAR system using GMM,

$$Y_{i,t} = \alpha + \sum_{j=1}^2 \beta_j Y_{i,t-j} + f_i + \varepsilon_{i,t} \quad (2)$$

where $Y_{i,t}$ denotes a six-variable vector (panel) of the following variables in order: (1) international industrial production growth, (2) change in the international interest rate, (3) U.S. bond market volatility, (4) U.S. stock market volatility, (5) international bond market volatility, and (6) international stock market volatility.¹⁷ We also include country fixed effects (f_i) to control

¹⁷These six variables (panel) are the same variables that we considered in our benchmark dynamic panel specification in Section 3.2. We first consider bonds with a maturity of one year or more. For robustness, we also consider a four-variable panel VAR specification. These results are reported in Appendix E.

for heterogeneity arising from a country’s individual characteristics. Because the fixed effects are usually correlated with the regressors as noted by Love and Zicchino (2006), we use a forward mean-differencing procedure and use lagged regressors as instruments for our GMM estimation.¹⁸

To provide structure for our panel VAR exercise, we use the Cholesky decomposition with a lower triangular matrix restriction. Our ordering of the variables, denoted above, implies that a shock to international industrial production growth has an immediate impact on U.S. and international financial market volatility, but not the other way around. In other words, variables that appear later in the ordering are assumed to affect variables that appear earlier only with a lag. For robustness, we consider alternative orderings of the four financial market volatility variables and we find that the results are qualitatively consistent with the benchmark ordering. To economize on space, we do not report these results in the paper.

An alternative approach to our panel VAR structure would be to use a proxy VAR as in Mertens and Ravn (2012) and Piffer and Podstawski (2018), with the use of exogenous variables in the spirit of Romer and Romer (2010) (that is, variables that are ‘not motivated by current or projected economic conditions’). Although the proxy VAR approach has some advantages, implementing such a method in our panel setting is challenging given data issues for a couple of reasons. First, proxy measures of both bond and stock market volatility that are methodologically consistent across countries would need to be available. Second, since our investigation concerns both bond and stock markets, it would be a challenge to identify exogenous sources of volatility that are specific to the two markets individually for each country. For example, under the narrative approach of Mertens and Ravn (2012), any announcement that potentially results in changes in taxes would likely impact both the bond and stock markets and it would be difficult to precisely identify the extent to which such an announcement impacts each of the two markets separately.

As a result, our analysis follows the recursive ordering approach, where our structure is motivated, in part, by the observation that output takes time to fully respond to shocks because

¹⁸We use up to 6 lags as instruments for our GMM estimation, but our generated impulse response functions generally are not sensitive to a larger set of instruments.

firms are constrained in terms of the effective timing of their investment and construction of productive capital. Examples of channels through which this can occur include, but are not limited to, the time-to-build channel (e.g., Kydland and Prescott (1982)), the capital accumulation and utilization channel (e.g., Greenwood et al. (1988)), the inventory and delivery channel (e.g., Khan and Thomas (2007)), and the information inattentiveness by firms channel (e.g., Reis (2006)). We also observe that financial markets respond almost immediately to news regarding changes in industrial production growth and policy rates (e.g., Boyd et al. (2005)), while industrial production growth and policy rates do not necessarily respond immediately to fluctuations in the financial market.¹⁹

4.2 IRFs and Variance Decomposition

Figure 4 presents the impulse response functions of international industrial production growth to the U.S. and international financial market volatility shocks for up to ten months after the shocks from the six-variable panel structural VAR where we include 95% confidence bands from 1,000 Monte Carlo simulations.²⁰ The impulse responses of international industrial production growth to one-standard-deviation increases in U.S. financial market volatility are the top sub-figures and the impulse responses to one-standard-deviation increases in international financial market volatility are the bottom sub-figures. Since we are controlling for country-specific heterogeneity in the panel structural VAR, these impulse response functions can be interpreted as the average responses of international industrial production growth to unexpected one-standard-deviation increases in U.S. and international financial market volatility.

[Figure 4 Here]

Consistent with our previous dynamic panel regression results, unexpected one-standard-deviation increases in U.S. financial market volatility imply negative spillover effects on international industrial production growth over time. Also, changes in U.S. financial market conditions appear to be more important for understanding international output growth than changes in

¹⁹Note that for many countries, short-term interest rates are highly correlated with policy rates.

²⁰Our results are robust to a higher number (i.e., 4,000) of Monte Carlo simulations.

a country's own financial market conditions. Specifically, international industrial production growth appears to respond more negatively to unexpected one-standard-deviation shocks arising from U.S. financial market volatility than to international financial market volatility shocks. Moreover, the impacts of U.S. financial market volatility shocks are also more persistent and become increasingly more negative over the horizon.

Building on the implications of these impulse response functions, in Figure 5, we decompose the variance contribution of each financial market volatility variable for international industrial production growth. We present the contributions of these four volatility measures for up to ten months after the initial shocks. Here, the black bars represent U.S. bond market volatility, the grey bars represent U.S. stock market volatility, the blue bars represent international bond market volatility, and the red bars represent international stock market volatility.

[Figure 5 Here]

As shown in Figure 5, we find that U.S. bond market volatility is able to explain a larger share of variation in international industrial production growth than are U.S. stock market volatility, international stock market volatility, and international bond market volatility from four to ten months after the shocks. On the other hand, U.S. stock market volatility, more often than not, accounts for a larger share of the variation in industrial production growth than international stock market volatility. These results continue to highlight the importance of understanding U.S. financial market volatility for explaining movements in international output growth and are consistent with our earlier exposition from the dynamic panel regression analysis in Section 3.

4.3 Term Structure Analysis

We next consider the corresponding impulse response analysis and variance decomposition analysis from Section 4.2 with different bond maturities (three, five, seven, and ten years). Figure 6 plots the impulse response functions of international industrial production growth to unexpected one-standard-deviation shocks to our measures of U.S. and international financial market volatil-

ity (columns) using bonds with the different maturities (rows). We find that the volatility of U.S. bonds with longer maturities tends to exhibit a more pronounced impact on international industrial production growth than the other financial market volatility variables. Also, U.S. stock market volatility appears to be more important in explaining variation in international output growth than U.S. bond market volatility that is constructed from bonds with maturities of one to three years (three).

[Figure 6 Here]

Figure 7 presents the variance decomposition of industrial production growth where we consider different bond maturities (three, five, seven, and ten years). Again, we find that the volatility of U.S. bonds with longer maturities is able to explain more variation in international industrial production growth than the other financial market volatility measures. Moreover, U.S. stock market volatility accounts for a large share of the variation in international output growth when bonds have a maturity of one to three years (three).

[Figure 7 Here]

Overall, these impulse response functions and variance decomposition results suggest that there are significant spillover effects of U.S. financial volatility on international output growth, which we also found in the dynamic panel regression analysis. In particular, the volatility of U.S. bonds with longer maturities is more useful for understanding international industrial production growth than volatility of U.S. bonds with shorter maturities. Also, U.S. stock market volatility is better positioned to explain movements in international output growth than U.S. bond market volatility when the bonds have shorter maturities.

5 Conclusion

This paper finds that there are statistically significant spillover effects of U.S. financial market volatility on international industrial production growth using a panel dataset of seventeen developed countries. In particular, times of high U.S. bond market volatility are associated with

low international output growth, volatility of U.S. bonds with longer maturities has a more statistically significant impact on international industrial production growth than volatility of U.S. bonds with shorter maturities, and U.S. financial market volatility plays a larger role statistically in explaining international output growth than a country's own financial market volatility. As a consequence of financial globalization, these results suggest that a country's macroeconomic condition is sensitive to external financial market conditions.

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Table 1: Bond and Stock Market Volatility

Country	Stats	Bond Volatility by Maturity					Stock Vol
		Three	Five	Seven	Ten	One Plus	
Austria	Mean	0.0050	0.0052	0.0054	0.0057	0.0055	0.0125
	Std	0.0017	0.0017	0.0018	0.0018	0.0017	0.0074
Belgium	Mean	0.0051	0.0053	0.0056	0.0059	0.0056	0.0110
	Std	0.0017	0.0018	0.0019	0.0020	0.0018	0.0058
Canada	Mean	0.0043	0.0044	0.0046	0.0049	0.0046	0.0104
	Std	0.0019	0.0019	0.0018	0.0018	0.0018	0.0060
Denmark	Mean	0.0050	0.0051	0.0053	0.0055	0.0053	0.0109
	Std	0.0016	0.0017	0.0017	0.0018	0.0017	0.0054
Finland	Mean	0.0053	0.0052	0.0058	0.0056	0.0053	0.0163
	Std	0.0035	0.0029	0.0061	0.0018	0.0017	0.0086
France	Mean	0.0050	0.0052	0.0054	0.0057	0.0055	0.0118
	Std	0.0017	0.0017	0.0018	0.0018	0.0018	0.0061
Germany	Mean	0.0050	0.0051	0.0053	0.0056	0.0053	0.0123
	Std	0.0016	0.0017	0.0018	0.0018	0.0017	0.0061
Ireland	Mean	0.0056	0.0061	0.0053	0.0064	0.0060	0.0130
	Std	0.0083	0.0105	0.0065	0.0064	0.0026	0.0075
Italy	Mean	0.0052	0.0056	0.0059	0.0062	0.0059	0.0126
	Std	0.0018	0.0021	0.0023	0.0025	0.0022	0.0067
Japan	Mean	0.0054	0.0055	0.0056	0.0058	0.0056	0.0113
	Std	0.0021	0.0022	0.0022	0.0022	0.0022	0.0047
Netherlands	Mean	0.0050	0.0052	0.0054	0.0056	0.0054	0.0111
	Std	0.0016	0.0017	0.0018	0.0018	0.0017	0.0061
Norway	Mean	0.0063	0.0062	0.0061	0.0063	0.0061	0.0136
	Std	0.0055	0.0056	0.0024	0.0024	0.0023	0.0079
Portugal	Mean	0.0055	0.0061	0.0067	0.0071	0.0064	0.0110
	Std	0.0024	0.0032	0.0038	0.0038	0.0031	0.0051
Spain	Mean	0.0052	0.0056	0.0059	0.0063	0.0059	0.0129
	Std	0.0018	0.002	0.0022	0.0024	0.0021	0.0066
Sweden	Mean	0.0058	0.0059	0.0067	0.0062	0.0060	0.0141
	Std	0.0022	0.0022	0.0112	0.0022	0.0022	0.0074
United Kingdom	Mean	0.0045	0.0046	0.0048	0.0027	0.0053	0.0101
	Std	0.0019	0.0019	0.0019	0.0009	0.0019	0.0058
United States	Mean	0.0006	0.0017	0.0024	0.0032	0.0023	0.0089
	Std	0.0004	0.0007	0.0009	0.0011	0.0008	0.0053
Total (All)	Mean	0.0049	0.0052	0.0054	0.0056	0.0054	0.0120
	Std	0.0032	0.0036	0.0041	0.0028	0.0022	0.0067

Note: This table presents the mean and standard deviation (std) of realized bond and stock market volatility by country. We consider bonds with maturities of one to three years (three), three to five years (five), five to seven years (seven), seven to ten years (ten), and one or more years (one plus). The time period is 1998M3–2017M2.

Table 2: Dynamic Panel Regressions (Bond Maturity - One Plus)

	Dependent Variable: IP Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
US Bond Vol (1-lag)	-0.683*** (-2.648)		-0.846*** (-2.883)	-0.710** (-2.390)	-0.757** (-2.536)	-0.515* (-1.719)
US Bond Vol (2-lag)	-1.914*** (-7.422)		-1.213*** (-3.907)	-1.148*** (-3.640)	-1.114*** (-3.519)	-1.126*** (-3.560)
US Stock Vol (1-lag)		0.054 (1.254)	0.138*** (2.888)	0.150*** (3.123)	0.193*** (3.092)	0.173*** (2.784)
US Stock Vol (2-lag)		-0.370*** (-8.667)	-0.249*** (-5.030)	-0.231*** (-4.596)	-0.158*** (-2.580)	-0.101 (-1.619)
IP Growth (1-lag)	-0.098*** (-6.182)	-0.091*** (-5.754)	-0.103*** (-6.455)	-0.103*** (-6.452)	-0.105*** (-6.578)	-0.121*** (-7.609)
IP Growth (2-lag)	0.073*** (4.298)	0.083*** (4.776)	0.072*** (4.223)	0.070*** (4.123)	0.069*** (4.076)	0.051*** (2.990)
Bond Vol (1-lag)				-0.197* (-1.921)	-0.166 (-1.508)	-0.129 (-1.176)
Bond Vol (2-lag)				-0.058 (-0.569)	0.009 (0.086)	0.041 (0.380)
Stock Vol (1-lag)					-0.043 (-0.889)	-0.049 (-0.998)
Stock Vol (2-lag)					-0.095* (-1.960)	-0.070 (-1.445)
Δ Interest Rate (1-lag)						0.433*** (4.045)
Δ Interest Rate (2-lag)						0.348*** (3.225)
Observations	3,600	3,600	3,600	3,600	3,600	3,600
R^2	0.050	0.045	0.056	0.058	0.058	0.068

Note: This table presents the results from a least square dummy variable (LSDV) regression with HAC standard errors (Equation (1)). We select 2 lags for each country based on the SBIC. We consider bonds with maturities of one or more years (one plus). The time period is 1998M3-2017M2. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Dynamic Panel Regressions with Additional Controls (Bond Maturity - One Plus)

	Dependent Variable: IP Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
US Bond Vol (1-lag)	-0.531* (-1.657)	-0.596* (-1.865)	-0.502 (-1.575)	-0.518 (-1.617)	-0.437 (-1.343)	-0.469 (-1.033)
US Bond Vol (2-lag)	-1.033*** (-3.040)	-0.987*** (-2.910)	-0.909*** (-2.678)	-0.917*** (-2.685)	-0.859** (-2.488)	-0.972** (-2.017)
US Stock Vol (1-lag)	0.182*** (2.733)	0.155** (2.294)	0.109 (1.595)	0.108 (1.570)	0.098 (1.414)	0.085 (0.870)
US Stock Vol (2-lag)	-0.080 (-1.204)	-0.087 (-1.311)	-0.101 (-1.522)	-0.094 (-1.392)	-0.095 (-1.397)	-0.107 (-1.127)
IP Growth (1-lag)	-0.121*** (-7.573)	-0.118*** (-7.444)	-0.130*** (-8.178)	-0.132*** (-8.310)	-0.137*** (-8.621)	-0.139*** (-8.713)
IP Growth (2-lag)	0.052*** (2.825)	0.053*** (2.914)	0.038** (2.071)	0.041** (2.184)	0.037* (1.933)	0.037 (1.405)
Bond Vol (1-lag)	-0.135 (-1.152)	-0.149 (-1.272)	-0.162 (-1.379)	-0.167 (-1.413)	-0.175 (-1.470)	-0.177 (-1.062)
Bond Vol (2-lag)	0.042 (0.360)	0.029 (0.252)	0.051 (0.443)	0.056 (0.482)	0.045 (0.387)	0.050 (0.306)
Stock Vol (1-lag)	-0.075 (-1.421)	-0.065 (-1.231)	-0.077 (-1.452)	-0.076 (-1.433)	-0.063 (-1.169)	-0.077 (-1.021)
Stock Vol (2-lag)	-0.081 (-1.554)	-0.084 (-1.614)	-0.082 (-1.587)	-0.082 (-1.568)	-0.077 (-1.454)	-0.078 (-1.058)
Δ Interest Rate (1-lag)	0.456*** (3.982)	0.471*** (4.106)	0.476*** (4.158)	0.473*** (4.105)	0.494*** (4.259)	0.461*** (2.837)
Δ Interest Rate (2-lag)	0.375*** (3.252)	0.376*** (3.259)	0.333*** (2.879)	0.324*** (2.786)	0.338*** (2.871)	0.334** (2.031)
EPU	0.916*** (3.266)	0.247 (0.613)	0.400 (0.988)	0.476 (1.162)	0.540 (1.301)	1.190* (1.899)
US EPU		1.361** (2.347)	0.796 (1.347)	0.635 (1.049)	0.636 (1.041)	-0.134 (-0.150)
Δ CLI			-4.357*** (-4.866)	-4.237*** (-4.689)	-4.827*** (-5.108)	-5.670*** (-4.203)
Δ MCI				-0.064 (-1.442)	-0.067 (-1.496)	-0.071 (-1.139)
Oil Volatility					-0.308** (-2.142)	-0.054 (-0.248)
US IP Growth						-0.018*** (-2.985)
Observations	3,600	3,600	3,600	3,600	3,600	3,600
R^2	0.071	0.073	0.079	0.079	0.081	0.086

Note: See notes to Table 2. EPU and US EPU are economic policy uncertainty indices by Baker et al. (2016), Δ CLI is the change in the composite leading indicator, Δ MCI is the change in Michigan's Consumer Sentiment Index, and oil volatility is from West Texas Intermediate (WTI). t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Dynamic Panel Regressions with Different Bond Maturities

	Dependent Variable: IP Growth				
	Bond Maturity				
	Three	Five	Seven	Ten	One Plus
US Bond Vol (1-lag)	0.305 (0.459)	-0.440 (-1.315)	-0.617** (-2.302)	-0.629*** (-2.902)	-0.515* (-1.719)
US Bond Vol (2-lag)	-1.070 (-1.574)	-0.633* (-1.821)	-0.944*** (-3.370)	-0.933*** (-4.074)	-1.126*** (-3.560)
US Stock Vol (1-lag)	0.144** (2.342)	0.181*** (2.881)	0.190*** (3.079)	0.175*** (2.847)	0.173*** (2.784)
US Stock Vol (2-lag)	-0.175*** (-2.934)	-0.157** (-2.550)	-0.128** (-2.106)	-0.101* (-1.648)	-0.101 (-1.619)
IP Growth (1-lag)	-0.114*** (-7.190)	-0.118*** (-7.397)	-0.123*** (-7.723)	-0.127*** (-7.996)	-0.121*** (-7.609)
IP Growth (2-lag)	0.059*** (3.395)	0.055*** (3.170)	0.052*** (3.005)	0.045*** (2.645)	0.051*** (2.990)
Bond Vol (1-lag)	-0.062 (-1.063)	-0.031 (-0.629)	0.014 (0.331)	-0.048 (-0.674)	-0.129 (-1.176)
Bond Vol (2-lag)	0.021 (0.382)	-0.024 (-0.493)	0.002 (0.059)	0.066 (0.911)	0.041 (0.380)
Stock Vol (1-lag)	-0.081* (-1.740)	-0.078* (-1.689)	-0.061 (-1.329)	-0.039 (-0.834)	-0.049 (-0.998)
Stock Vol (2-lag)	-0.065 (-1.368)	-0.068 (-1.437)	-0.060 (-1.297)	-0.056 (-1.203)	-0.070 (-1.445)
Δ Interest Rate (1-lag)	0.432*** (3.986)	0.434*** (4.034)	0.441*** (4.119)	0.428*** (4.018)	0.433*** (4.045)
Δ Interest Rate (2-lag)	0.435*** (4.055)	0.414*** (3.879)	0.392*** (3.683)	0.337*** (3.164)	0.348*** (3.225)
Observations	3,600	3,600	3,600	3,600	3,600
R^2	0.061	0.063	0.068	0.073	0.068

Note: See notes to Table 2. We consider bonds with maturities of one to three years (three), three to five years (five), five to seven years (seven), seven to ten years (ten), and one or more years (one plus). t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Pre-Great Recession Dynamic Panel Regressions with Different Bond Maturities

	Dependent Variable: IP Growth				
	Bond Maturity				
	Three	Five	Seven	Ten	One Plus
US Bond Vol (1-lag)	-1.733* (-1.725)	-1.330*** (-2.702)	-1.093*** (-2.708)	-1.028*** (-3.203)	-1.566*** (-3.621)
US Bond Vol (2-lag)	-1.885* (-1.862)	-0.679 (-1.393)	-0.972** (-2.456)	-0.643** (-2.032)	-1.281*** (-3.074)
US Stock Vol (1-lag)	0.188** (2.034)	0.211** (2.261)	0.186** (2.042)	0.187** (2.073)	0.192** (2.164)
US Stock Vol (2-lag)	0.095 (1.002)	0.105 (1.093)	0.176* (1.826)	0.158* (1.663)	0.240** (2.536)
IP Growth (1-lag)	-0.325*** (-14.941)	-0.330*** (-15.192)	-0.330*** (-15.244)	-0.328*** (-15.144)	-0.331*** (-15.306)
IP Growth (2-lag)	-0.161*** (-7.007)	-0.166*** (-7.224)	-0.162*** (-7.127)	-0.163*** (-7.173)	-0.173*** (-7.669)
Bond Vol (1-lag)	-0.038 (-0.458)	0.028 (0.366)	0.025 (0.576)	-0.039 (-0.450)	-0.291* (-1.728)
Bond Vol (2-lag)	0.086 (1.042)	0.072 (0.936)	0.037 (0.832)	0.143* (1.648)	0.465*** (2.687)
Stock Vol (1-lag)	0.028 (0.419)	0.016 (0.242)	0.017 (0.254)	0.019 (0.278)	0.046 (0.672)
Stock Vol (2-lag)	-0.058 (-0.870)	-0.067 (-1.004)	-0.059 (-0.883)	-0.071 (-1.072)	-0.110* (-1.646)
Δ Interest Rate (1-lag)	0.005*** (3.394)	0.005*** (3.230)	0.004*** (3.219)	0.004*** (3.245)	0.005*** (3.306)
Δ Interest Rate (2-lag)	0.003** (2.248)	0.003** (2.052)	0.003* (1.842)	0.002* (1.702)	0.001 (0.842)
Observations	1,824	1,824	1,824	1,824	1,824
R^2	0.078	0.080	0.083	0.085	0.093

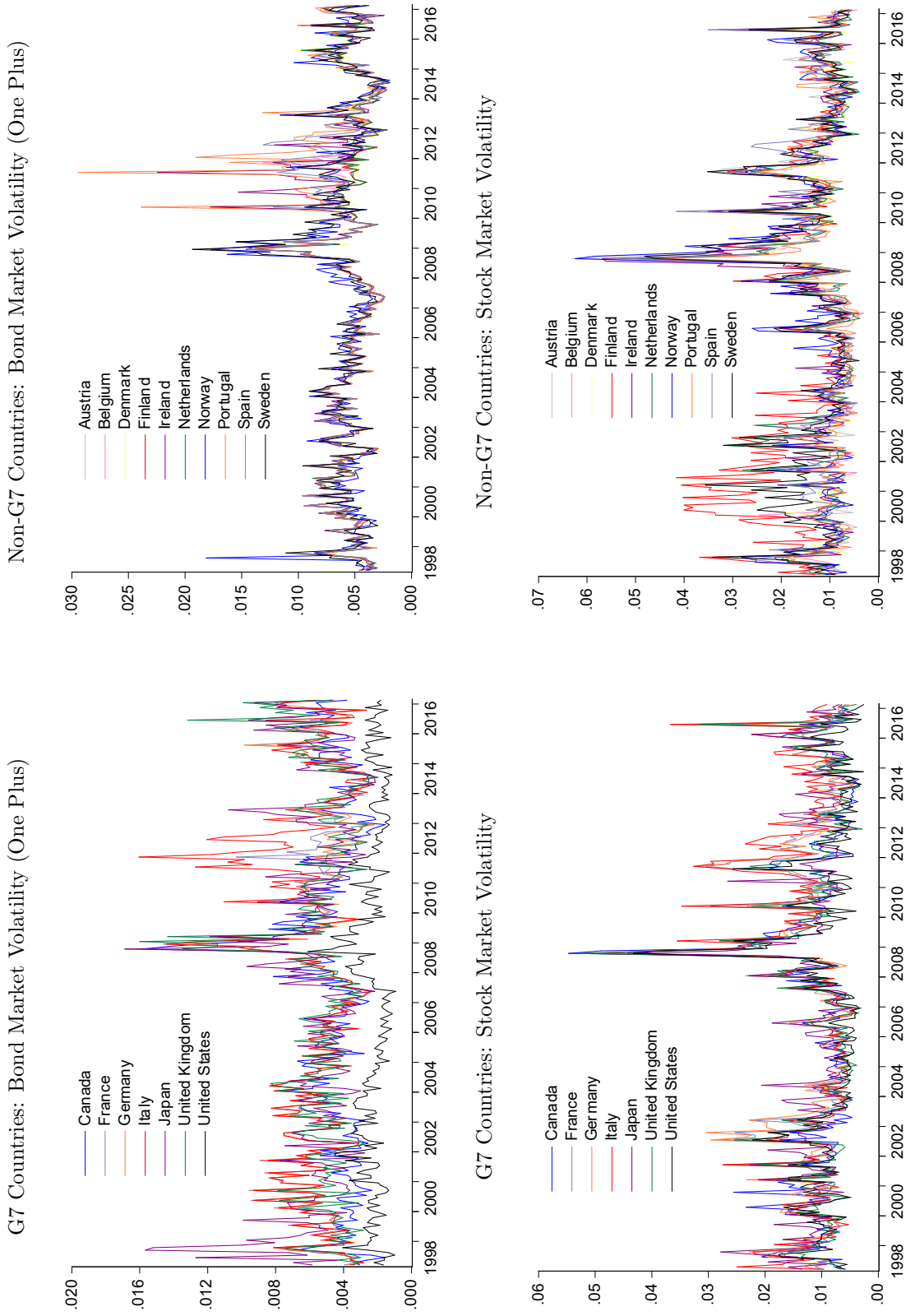
Note: See notes to Table 2. We consider bonds with maturities of one to three years (three), three to five years (five), five to seven years (seven), seven to ten years (ten), and one or more years (one plus). The time period is 1998M3-2007M11. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: G7 vs. non-G7 Dynamic Panel Regressions with Different Bond Maturities

	Dependent Variable: IP Growth											
	Bond Maturity											
	Three		Five		Seven		Ten		One Plus			
	G7	Non-G7	G7	Non-G7	G7	Non-G7	G7	Non-G7	G7	Non-G7	G7	Non-G7
US Bond Vol (1-lag)	0.765 (0.663)	-0.099 (-0.128)	-0.132 (-0.222)	-0.646 (-1.626)	-0.238 (-0.493)	-0.834*** (-2.587)	-0.380 (-0.946)	-0.789*** (-2.992)	-0.380 (-0.946)	-0.789*** (-2.992)	-0.380 (-0.946)	-0.789*** (-2.992)
US Bond Vol (2-lag)	-2.286** (-1.979)	-0.198 (-0.246)	-1.260** (-2.121)	-0.168 (-0.410)	-1.421*** (-2.911)	-0.546* (-1.655)	-1.248*** (-3.107)	-0.642** (-2.385)	-1.248*** (-3.107)	-0.642** (-2.385)	-1.248*** (-3.107)	-0.642** (-2.385)
US Stock Vol (1-lag)	0.087 (0.759)	0.173** (2.372)	0.105 (0.895)	0.214*** (2.844)	0.092 (0.795)	0.234*** (3.119)	0.072 (0.626)	0.222*** (2.974)	0.072 (0.626)	0.222*** (2.974)	0.072 (0.626)	0.222*** (2.974)
US Stock Vol (2-lag)	-0.147 (-1.307)	-0.196*** (-2.692)	-0.118 (-1.026)	-0.182** (-2.453)	-0.086 (-0.752)	-0.155** (-2.109)	-0.065 (-0.565)	-0.128* (-1.743)	-0.065 (-0.565)	-0.128* (-1.743)	-0.065 (-0.565)	-0.128* (-1.743)
IP Growth (1-lag)	-0.239*** (-9.049)	-0.031 (-1.504)	-0.242*** (-9.184)	-0.035* (-1.666)	-0.246*** (-9.344)	-0.040* (-1.903)	-0.252*** (-9.574)	-0.043** (-2.050)	-0.252*** (-9.574)	-0.043** (-2.050)	-0.252*** (-9.574)	-0.043** (-2.050)
IP Growth (2-lag)	-0.025 (-0.917)	0.108*** (4.913)	-0.030 (-1.075)	0.106*** (4.814)	-0.032 (-1.174)	0.101*** (4.646)	-0.038 (-1.403)	0.095*** (4.359)	-0.038 (-1.403)	0.095*** (4.359)	-0.038 (-1.403)	0.095*** (4.359)
Bond Vol (1-lag)	0.054 (0.221)	-0.066 (-1.160)	0.082 (0.360)	-0.026 (-0.533)	0.050 (0.231)	0.012 (0.291)	0.096 (0.441)	-0.068 (-0.911)	0.096 (0.441)	-0.068 (-0.911)	0.096 (0.441)	-0.068 (-0.911)
Bond Vol (2-lag)	-0.434* (-1.812)	0.066 (1.182)	-0.368 (-1.614)	0.001 (0.030)	-0.287 (-1.330)	0.017 (0.455)	-0.321 (-1.497)	0.131* (1.746)	-0.321 (-1.497)	0.131* (1.746)	-0.321 (-1.497)	0.131* (1.746)
Stock Vol (1-lag)	-0.036 (-0.367)	-0.111** (-2.075)	-0.031 (-0.324)	-0.108** (-2.038)	0.002 (0.017)	-0.094* (-1.768)	0.019 (0.207)	-0.072 (-1.369)	0.019 (0.207)	-0.072 (-1.369)	0.019 (0.207)	-0.072 (-1.369)
Stock Vol (2-lag)	-0.113 (-1.157)	-0.014 (-0.258)	-0.134 (-1.402)	-0.015 (-0.284)	-0.132 (-1.420)	-0.008 (-0.144)	-0.121 (-1.341)	-0.009 (-0.174)	-0.121 (-1.341)	-0.009 (-0.174)	-0.121 (-1.341)	-0.009 (-0.174)
Δ Interest Rate (1-lag)	0.006*** (3.068)	0.004*** (2.913)	0.006*** (3.070)	0.003*** (2.930)	0.007*** (3.100)	0.004*** (2.979)	0.006*** (3.074)	0.003*** (2.841)	0.006*** (3.074)	0.003*** (2.841)	0.006*** (3.074)	0.003*** (2.841)
Δ Interest Rate (2-lag)	0.004** (2.095)	0.004*** (3.561)	0.004** (1.988)	0.004*** (3.435)	0.004* (1.916)	0.004*** (3.263)	0.004* (1.732)	0.003*** (2.813)	0.004* (1.732)	0.003*** (2.813)	0.004* (1.732)	0.003*** (2.813)
Observations	1,350	2,250	1,350	2,250	1,350	2,250	1,350	2,250	1,350	2,250	1,350	2,250
R^2	0.109	0.065	0.111	0.066	0.115	0.070	0.119	0.075	0.119	0.075	0.119	0.075

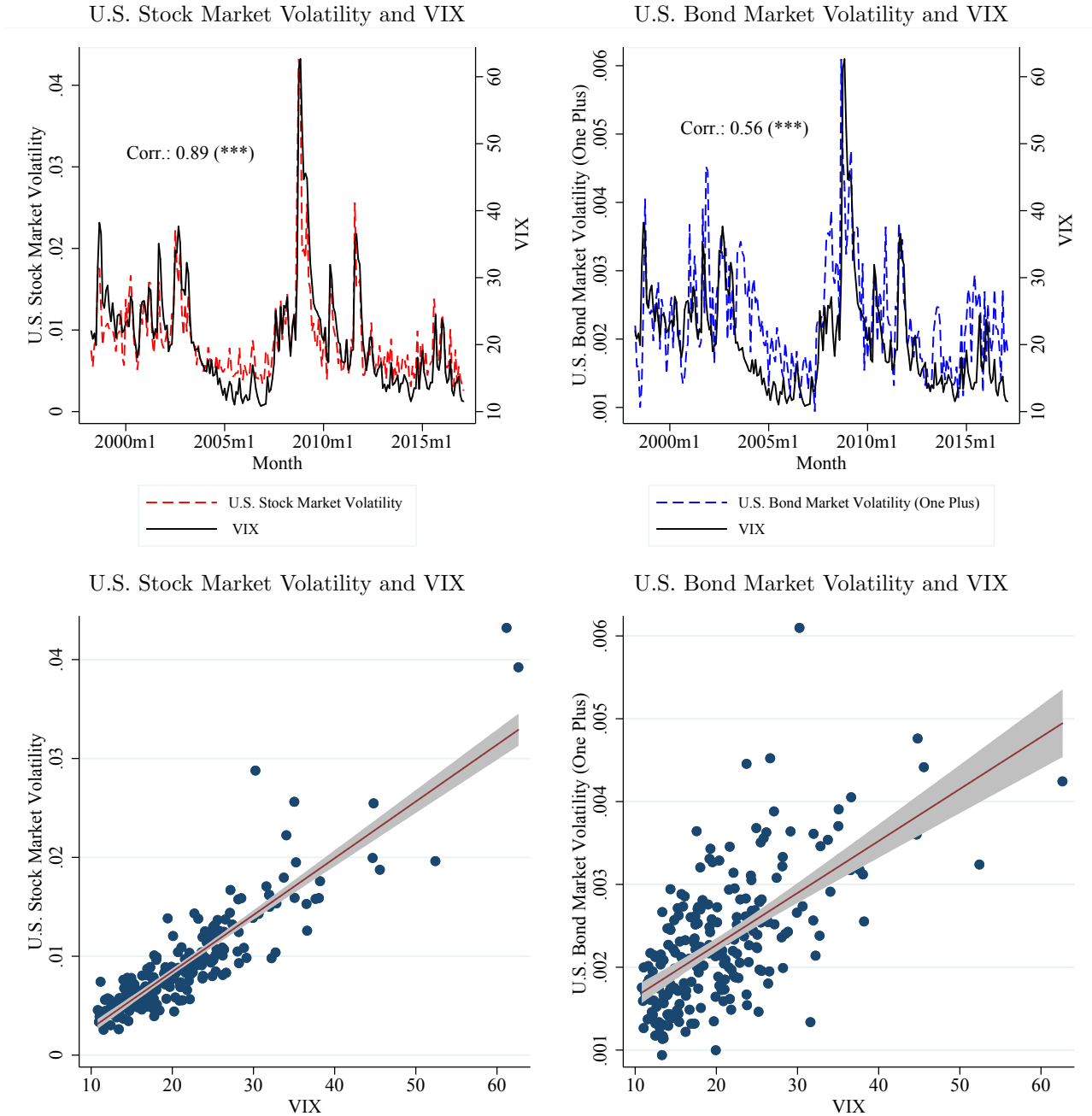
Note: See notes to Table 2. We consider bonds with maturities of one to three years (three), three to five years (five), five to seven years (seven), seven to ten years (ten), and one or more years (one plus). The panel sample is divided into G7 countries (excluding the United States) and non-G7 countries. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 1: Bond and Stock Market Volatility



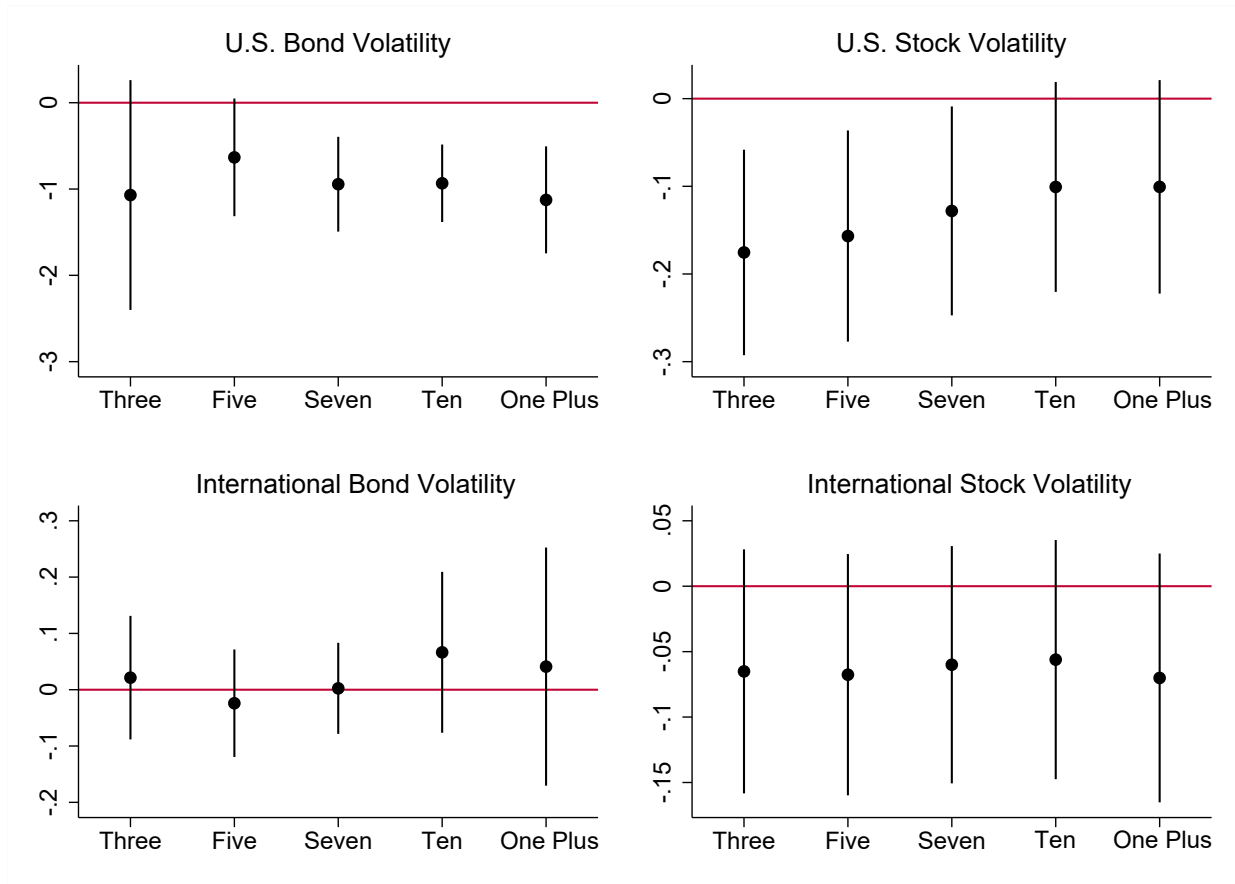
Note: This figure plots the realized country-specific bond and stock market volatility from 1998M3-2017M2. We consider bonds with maturities of one or more years (one plus).

Figure 2: U.S. Bond and Stock Market Volatility vs. VIX



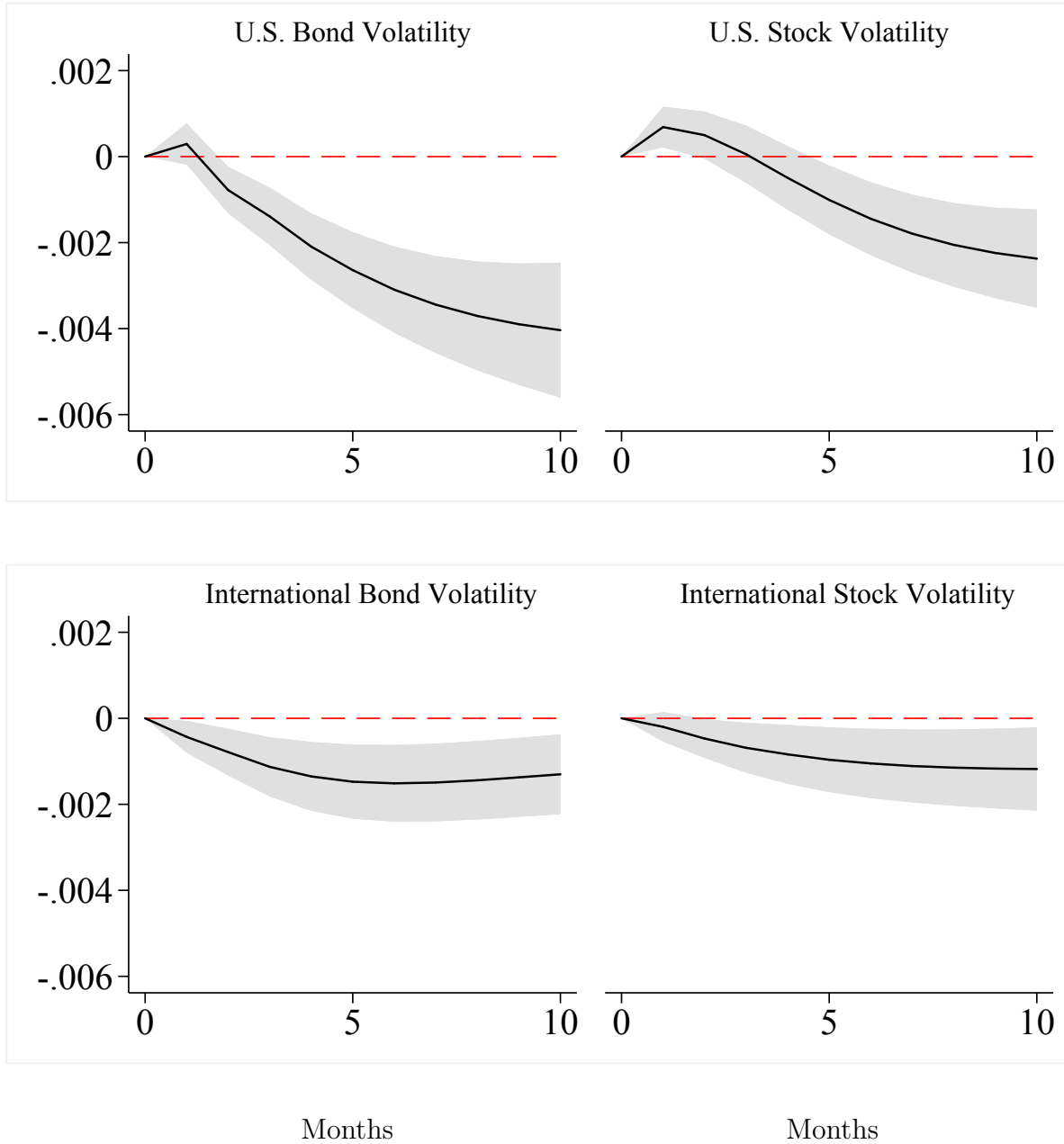
Note: This figure plots realized U.S. bond and stock market volatility against the Chicago Board Options Exchange (CBOE) Volatility Index (VIX), a measure of implied volatility. We consider bonds with maturities of one or more years (one plus). In the bottom two scatter plots, linear trends with the corresponding 95% confidence bands are included. The time period is 1998M3-2017M2.

Figure 3: Dynamic Panel Coefficients (Bond Maturities - Three, Five, Seven, Ten, One Plus)



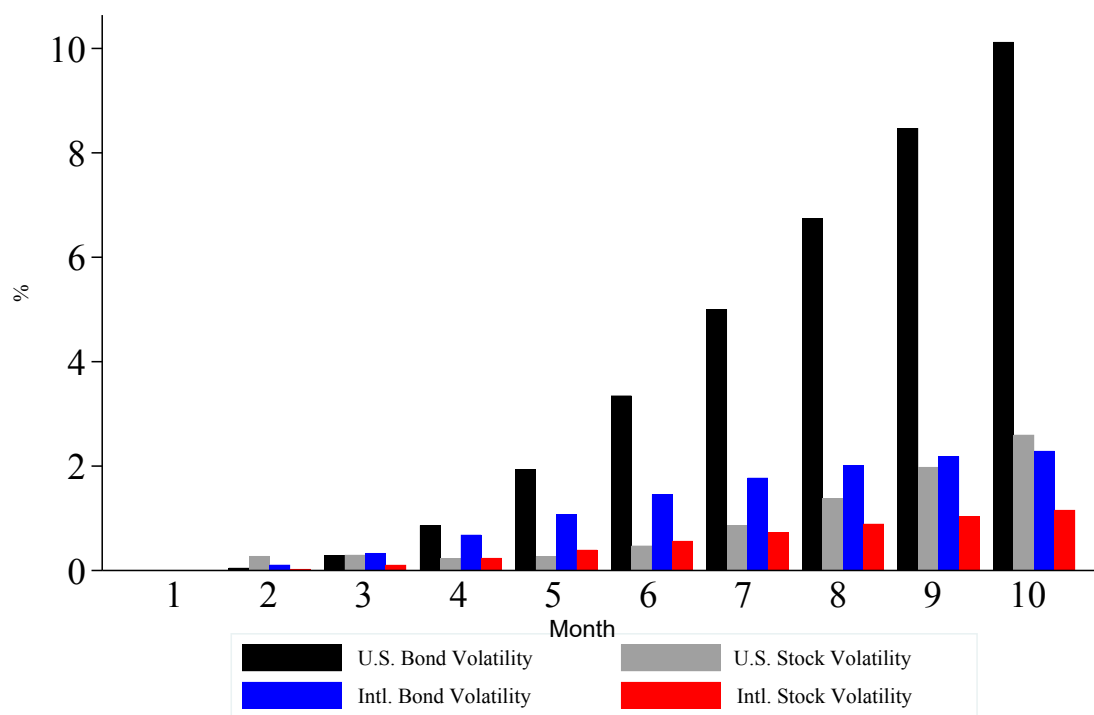
Note: This figure plots the second-lag coefficients and the corresponding confidence bands at the 5% significance level for U.S. bond volatility, U.S. stock volatility, international bond volatility, and international stock volatility from Table 4. We consider bonds with maturities of one to three years (three), three to five years (five), five to seven years (seven), seven to ten years (ten), and one or more years (one plus).

Figure 4: IRFs of International Industrial Production Growth to Financial Market Volatility Shocks (Bond Maturity - One Plus)



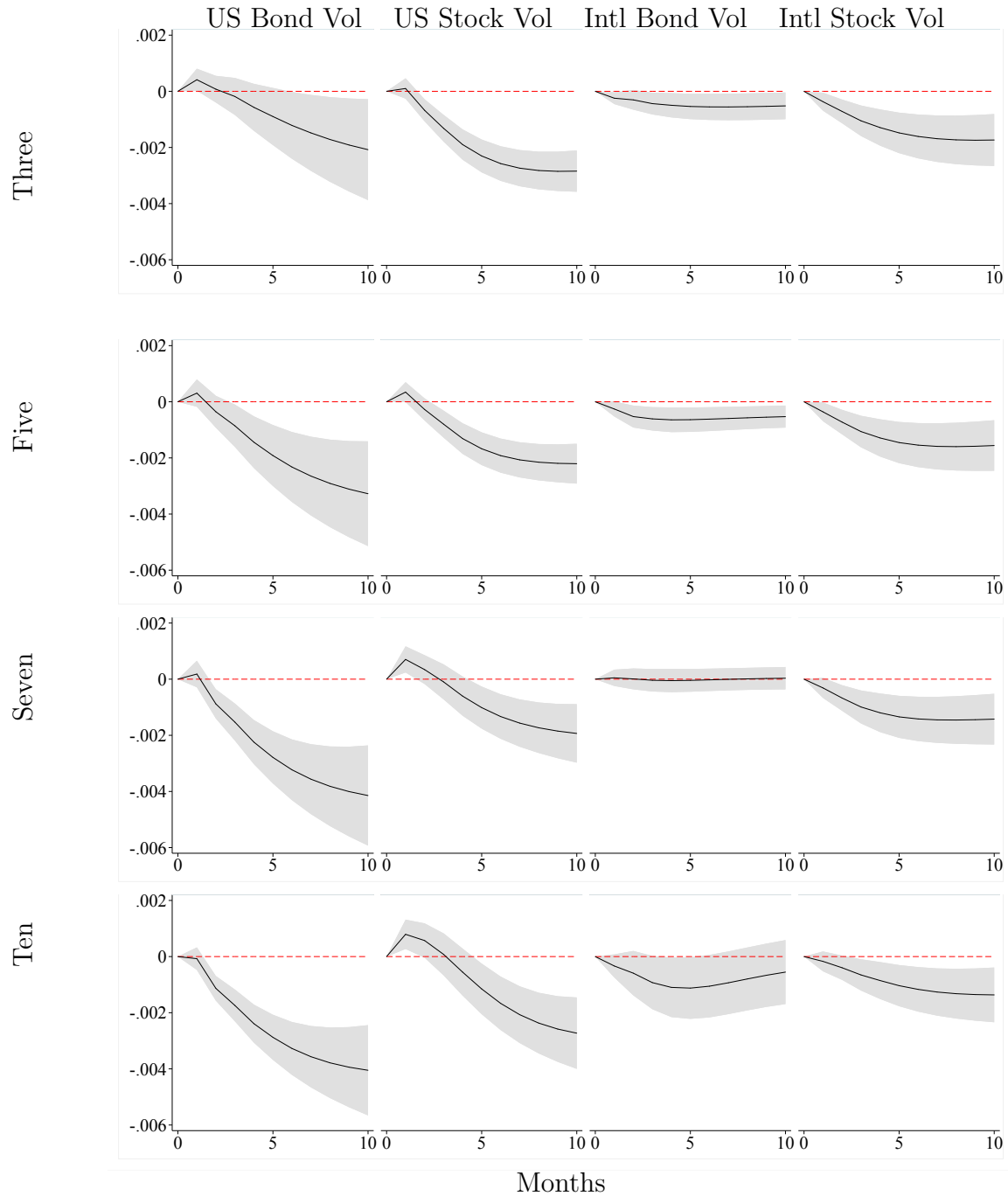
Note: This figure plots the impulse response functions of international industrial production growth to one-standard-deviation increases in the innovations of the financial market volatility variables for up to ten months after the shocks. These are generated from a six-variable panel VAR with the following variable order: international industrial production growth, international interest rate change, U.S. bond market volatility, U.S. stock market volatility, international bond market volatility, and international stock market volatility. We consider bonds with maturities of one or more years (one plus). 95% confidence bands from 1,000 Monte Carlo simulations are included in grey. The time period is 1998M3-2017M2.

Figure 5: Variance Decomposition of International Industrial Production Growth (Bond Maturity - One Plus)



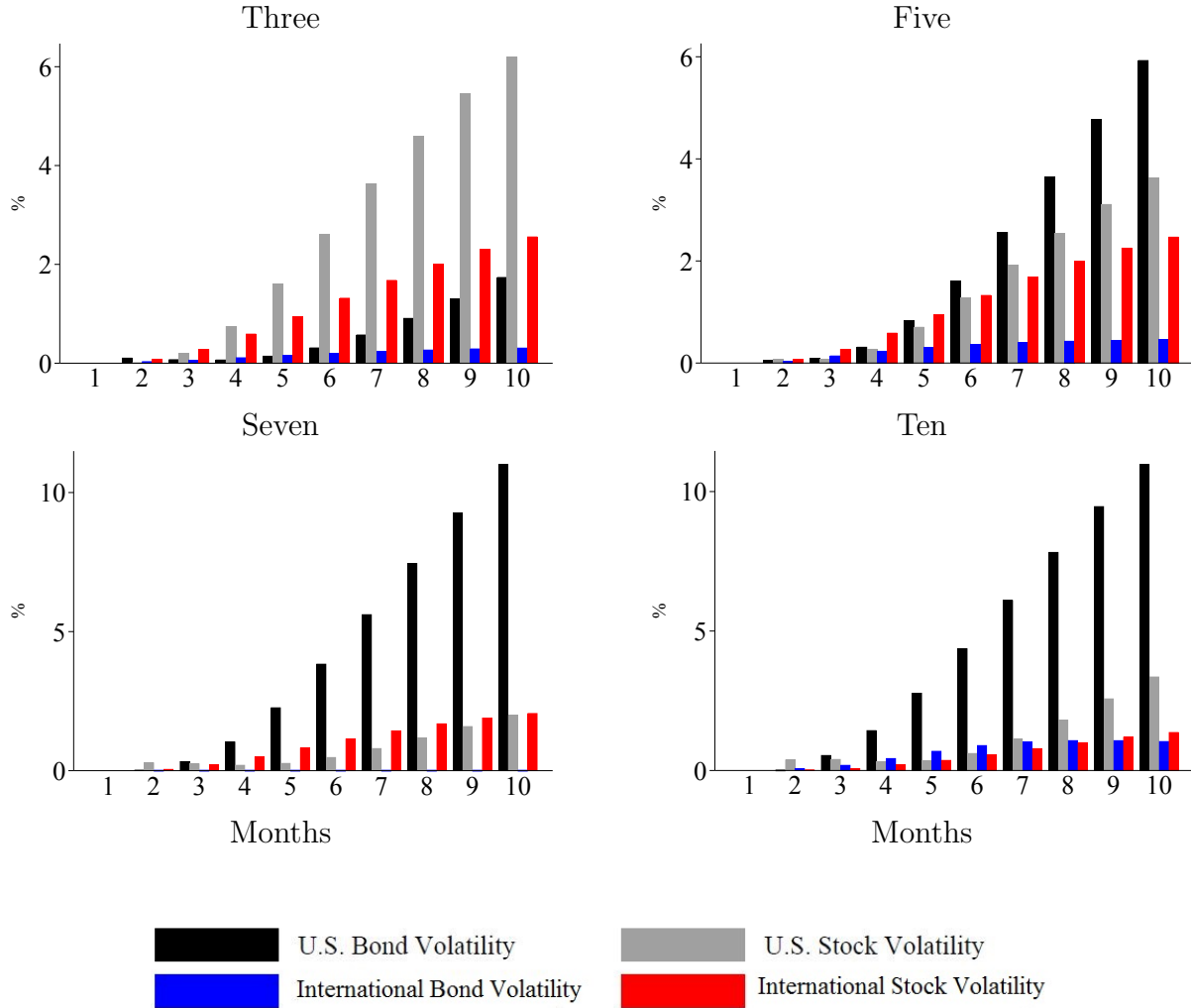
Note: This figure plots the variance decomposition of international industrial production growth for up to ten months after the shocks, which is generated from a six-variable panel VAR with the following variable order: international industrial production growth, international interest rate change, U.S. bond market volatility, U.S. stock market volatility, international bond market volatility, and international stock market volatility. We consider bonds with maturities of one or more years (one plus). The time period is 1998M3-2017M2.

Figure 6: IRFs of International Industrial Production Growth to Financial Market Volatility Shocks (Bond Maturities - Three, Five, Seven, Ten)



Note: This figure plots the impulse response functions of international industrial production growth to one-standard-deviation increases in the innovations of the financial market volatility variables for up to ten months after the shocks. These are generated from a six-variable panel VAR with the following variable order: international industrial production growth, international interest rate change, U.S. bond market volatility, U.S. stock market volatility, international bond market volatility, and international stock market volatility. We consider bonds with maturities of one to three years (three), three to five years (five), five to seven years (seven), and seven to ten years (ten). 95% confidence bands from 1,000 Monte Carlo simulations are included in grey. The time period is 1998M3-2017M2.

Figure 7: Variance Decomposition of International Industrial Production Growth (Bond Maturities - Three, Five, Seven, Ten)



Note: This figure plots the variance decomposition of international industrial production growth for up to ten months after the shocks, which is generated from a six-variable panel VAR with the following variable order: international industrial production growth, international interest rate change, U.S. bond market volatility, U.S. stock market volatility, international bond market volatility, and international stock market volatility. We consider bonds with maturities of one to three years (three), three to five years (five), five to seven years (seven), and seven to ten years (ten). The time period is 1998M3-2017M2.

Appendix (not intended for publication)

A Interest Rate Data

Table A.1: Short-term Interest Rates By Country

Austria	3-month vibor
Belgium	3-month treasury bill rate
Canada	3-month treasury bill
Denmark	3-month interbank offered rate
Finland	3-month helibor
France	3-month euribor
Germany	3-month fibor
Ireland	3-month interbank offered rate
Italy	3-month interbank deposit rate
Japan	3-month interbank rate
Netherlands	3-month interbank offered rate
Norway	3-month interbank rate
Portugal	3-month interbank offered rate
Spain	3-month interbank rate
Sweden	3-month interbank money rate
United Kingdom	3-month treasury bill rate
United States	3-month treasury bill rate

B Bond and Stock Market Indices

The raw bond market index data are the Bank of America Merrill Lynch (BofAML) bond index, provided by Bloomberg. The whole data set consists of 4,500 standard indices that track more than \$66 trillion of fixed income securities.²¹ Among these series, we focus on government (as opposed to corporate) bond indices. A more detailed construction of the index can be found in the BofAML bond index guide (Galdi et al. (2016)).

The raw stock market index data are the Morgan Stanley Capital International (MSCI) standard country equity indices that cover the majority of a country's stocks. The precise definition of the index is as follows²²

To construct a country index, every listed security in the market is identified. Securities are free float adjusted, classified in accordance with the Global Industry Classification Standard (GICS®), and screened by size, liquidity and minimum free float.

²¹From the 2016 BofAML bond index guide. Retrieved July 12, 2017.

²²Retrieved from Thompson Reuters' Datastream on July 12, 2017.

C Panel Unit Roots Test

As a preamble to any panel regression and VAR, we test for unit roots of our set of financial market volatility variables in a panel setting using the Im et al. (2003) (Im-Pesaran-Shin, or IPS) panel unit root test without time trends. As shown in Table C.2, we can reject the null hypothesis that all panels contain unit roots across all of our constructed measures of financial market volatility.

Table C.2: Im-Pesaran-Shin (IPS) Unit Root Test

Variable	$t - bar$	$\tilde{t} - bar$	$Z_{\tilde{t} - bar}$
Bond Market Volatility			
Three	-10.9883	-8.884	-38.6295
Five	-11.1868	-9.0022	-39.2488
Seven	-11.5186	-9.1903	-40.2345
Ten	-10.5638	-8.7238	-37.7898
One Plus	-10.4778	-8.648	-37.3928
Stock Market Volatility	-7.3495	-6.7008	-27.1875
H_o : All panels contain unit roots.			
H_a : Some panels are stationary.			

D Bond Market Volatility at Different Maturities

The dynamic panel regression results that correspond to Table 2 and Table 3 in the main text, but where we consider bonds with maturities of one to three years (three), three to five years (five), five to seven years (seven), and seven to ten years (ten), are presented in the tables below. These tables are meant to serve as robustness checks.

Table D.3: Dynamic Panel Regressions (Bond Maturity - Three)

	Dependent Variable: IP Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
US Bond Vol (1-lag)	0.666 (1.078)		0.475 (0.714)	0.482 (0.726)	0.220 (0.329)	0.305 (0.459)
US Bond Vol (2-lag)	-3.150*** (-5.111)		-0.835 (-1.213)	-0.889 (-1.293)	-0.882 (-1.291)	-1.070 (-1.574)
US Stock Vol (1-lag)		0.054 (1.254)	0.050 (1.055)	0.067 (1.412)	0.146** (2.362)	0.144** (2.342)
US Stock Vol (2-lag)		-0.370*** (-8.667)	-0.349*** (-7.484)	-0.322*** (-6.550)	-0.243*** (-4.109)	-0.175*** (-2.934)
IP Growth (1-lag)	-0.076*** (-4.749)	-0.091*** (-5.754)	-0.092*** (-5.790)	-0.093*** (-5.845)	-0.095*** (-5.981)	-0.114*** (-7.190)
IP Growth (2-lag)	0.094*** (5.443)	0.083*** (4.776)	0.084*** (4.850)	0.083*** (4.764)	0.081*** (4.681)	0.059*** (3.395)
Bond Vol (1-lag)				-0.131** (-2.276)	-0.101* (-1.715)	-0.062 (-1.063)
Bond Vol (2-lag)				-0.053 (-0.966)	-0.021 (-0.377)	0.021 (0.382)
Stock Vol (1-lag)					-0.078* (-1.653)	-0.081* (-1.740)
Stock Vol (2-lag)					-0.091* (-1.916)	-0.065 (-1.368)
Δ Interest Rate (1-lag)						0.432*** (3.986)
Δ Interest Rate (2-lag)						0.435*** (4.055)
Observations	3,600	3,600	3,600	3,600	3,600	3,600
R^2	0.026	0.045	0.046	0.048	0.049	0.061

Note: This table presents the results from a least square dummy variable (LSDV) regression with HAC standard errors (Equation (1)). We select 2 lags for each country based on the SBIC. We consider bonds with maturities of one to three years (three). The time period is 1998M3-2017M2. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.4: Dynamic Panel Regressions with Additional Controls (Bond Maturity - Three)

	Dependent Variable: IP Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
US Bond Vol (1-lag)	0.640 (0.895)	0.379 (0.521)	0.195 (0.268)	0.102 (0.138)	0.502 (0.663)	0.350 (0.337)
US Bond Vol (2-lag)	-0.459 (-0.601)	-0.408 (-0.536)	-0.727 (-0.957)	-0.710 (-0.932)	-0.725 (-0.940)	-0.725 (-0.683)
US Stock Vol (1-lag)	0.130* (1.955)	0.111* (1.664)	0.077 (1.155)	0.078 (1.162)	0.062 (0.907)	0.052 (0.553)
US Stock Vol (2-lag)	-0.165** (-2.554)	-0.170*** (-2.632)	-0.170*** (-2.646)	-0.164** (-2.541)	-0.160** (-2.442)	-0.179** (-1.983)
IP Growth (1-lag)	-0.114*** (-7.189)	-0.112*** (-7.074)	-0.126*** (-7.943)	-0.128*** (-8.055)	-0.134*** (-8.481)	-0.136*** (-8.525)
IP Growth (2-lag)	0.059*** (3.153)	0.060*** (3.227)	0.043** (2.293)	0.046** (2.391)	0.040** (2.073)	0.041 (1.530)
Bond Vol (1-lag)	-0.061 (-0.964)	-0.064 (-1.011)	-0.065 (-1.036)	-0.068 (-1.071)	-0.066 (-1.034)	-0.067 (-0.762)
Bond Vol (2-lag)	0.023 (0.373)	0.017 (0.287)	0.026 (0.434)	0.025 (0.419)	0.024 (0.400)	0.023 (0.280)
Stock Vol (1-lag)	-0.111** (-2.178)	-0.103** (-2.005)	-0.112** (-2.186)	-0.111** (-2.171)	-0.096* (-1.846)	-0.110 (-1.532)
Stock Vol (2-lag)	-0.069 (-1.345)	-0.075 (-1.451)	-0.075 (-1.455)	-0.074 (-1.445)	-0.065 (-1.254)	-0.067 (-0.940)
Δ Interest Rate (1-lag)	0.455*** (3.887)	0.468*** (4.001)	0.475*** (4.073)	0.473*** (4.036)	0.499*** (4.216)	0.469*** (2.872)
Δ Interest Rate (2-lag)	0.462*** (3.988)	0.463*** (4.005)	0.404*** (3.484)	0.396*** (3.404)	0.412*** (3.482)	0.411** (2.535)
EPU	1.029*** (3.353)	0.420 (0.944)	0.451 (1.016)	0.500 (1.121)	0.665 (1.462)	1.222* (1.819)
US EPU		1.168* (1.936)	0.616 (1.010)	0.486 (0.786)	0.432 (0.690)	-0.239 (-0.265)
Δ CLI			-4.833*** (-5.261)	-4.741*** (-5.136)	-5.398*** (-5.617)	-6.209*** (-4.597)
Δ MCI				-0.056 (-1.254)	-0.058 (-1.276)	-0.062 (-0.997)
Oil Volatility					-0.390*** (-2.656)	-0.163 (-0.741)
US IP Growth						-0.016*** (-2.678)
Observations	3,600	3,600	3,600	3,600	3,600	3,600
R^2	0.064	0.066	0.073	0.074	0.076	0.080

Note: See notes to Table D.3. EPU and US EPU are economic policy uncertainty indices by Baker et al. (2016), Δ CLI is the change in the composite leading indicator, Δ MCI is the change in Michigan's Consumer Sentiment Index, and oil volatility is from West Texas Intermediate (WTI). t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.5: Dynamic Panel Regressions (Bond Maturity - Five)

	Dependent Variable: IP Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
US Bond Vol (1-lag)	-0.352 (-1.186)		-0.553* (-1.654)	-0.516 (-1.544)	-0.617* (-1.843)	-0.440 (-1.315)
US Bond Vol (2-lag)	-1.717*** (-5.774)		-0.592* (-1.688)	-0.556 (-1.583)	-0.537 (-1.534)	-0.633* (-1.821)
US Stock Vol (1-lag)		0.054 (1.254)	0.111** (2.295)	0.116** (2.384)	0.195*** (3.086)	0.181*** (2.881)
US Stock Vol (2-lag)		-0.370*** (-8.667)	-0.333*** (-6.798)	-0.318*** (-6.387)	-0.230*** (-3.800)	-0.157** (-2.550)
IP Growth (1-lag)	-0.085*** (-5.340)	-0.091*** (-5.754)	-0.095*** (-5.949)	-0.096*** (-6.026)	-0.099*** (-6.202)	-0.118*** (-7.397)
IP Growth (2-lag)	0.083*** (4.854)	0.083*** (4.776)	0.079*** (4.603)	0.078*** (4.524)	0.076*** (4.426)	0.055*** (3.170)
Bond Vol (1-lag)				-0.062 (-1.275)	-0.035 (-0.716)	-0.031 (-0.629)
Bond Vol (2-lag)				-0.065 (-1.347)	-0.038 (-0.772)	-0.024 (-0.493)
Stock Vol (1-lag)					-0.079* (-1.696)	-0.078* (-1.689)
Stock Vol (2-lag)					-0.098** (-2.091)	-0.068 (-1.437)
Δ Interest Rate (1-lag)						0.434*** (4.034)
Δ Interest Rate (2-lag)						0.414*** (3.879)
Observations	3,600	3,600	3,600	3,600	3,600	3,600
R^2	0.035	0.045	0.049	0.049	0.051	0.063

Note: This table presents the results from a least square dummy variable (LSDV) regression with HAC standard errors (Equation (1)). We select 2 lags for each country based on the SBIC. We consider bonds with maturities of three to five years (five). The time period is 1998M3-2017M2. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.6: Dynamic Panel Regressions with Additional Controls (Bond Maturity - Five)

	Dependent Variable: IP Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
US Bond Vol (1-lag)	-0.347 (-0.958)	-0.487 (-1.333)	-0.450 (-1.237)	-0.493 (-1.345)	-0.355 (-0.944)	-0.398 (-0.772)
US Bond Vol (2-lag)	-0.438 (-1.137)	-0.393 (-1.024)	-0.427 (-1.112)	-0.419 (-1.087)	-0.345 (-0.882)	-0.420 (-0.785)
US Stock Vol (1-lag)	0.177*** (2.610)	0.154** (2.250)	0.108 (1.566)	0.108 (1.570)	0.094 (1.342)	0.083 (0.859)
US Stock Vol (2-lag)	-0.146** (-2.204)	-0.154** (-2.326)	-0.159** (-2.396)	-0.153** (-2.288)	-0.153** (-2.274)	-0.169* (-1.825)
IP Growth (1-lag)	-0.117*** (-7.342)	-0.115*** (-7.217)	-0.129*** (-8.097)	-0.131*** (-8.229)	-0.136*** (-8.539)	-0.137*** (-8.613)
IP Growth (2-lag)	0.056*** (2.984)	0.057*** (3.064)	0.040** (2.119)	0.042** (2.230)	0.038** (1.976)	0.039 (1.463)
Bond Vol (1-lag)	-0.033 (-0.620)	-0.038 (-0.730)	-0.051 (-0.970)	-0.051 (-0.968)	-0.054 (-1.014)	-0.057 (-0.777)
Bond Vol (2-lag)	-0.026 (-0.496)	-0.030 (-0.576)	-0.037 (-0.707)	-0.038 (-0.716)	-0.041 (-0.767)	-0.045 (-0.613)
Stock Vol (1-lag)	-0.103** (-2.031)	-0.093* (-1.829)	-0.103** (-2.035)	-0.103** (-2.027)	-0.091* (-1.764)	-0.105 (-1.481)
Stock Vol (2-lag)	-0.074 (-1.449)	-0.079 (-1.552)	-0.073 (-1.442)	-0.072 (-1.419)	-0.066 (-1.282)	-0.067 (-0.946)
Δ Interest Rate (1-lag)	0.454*** (3.897)	0.470*** (4.036)	0.475*** (4.087)	0.472*** (4.040)	0.494*** (4.189)	0.462*** (2.846)
Δ Interest Rate (2-lag)	0.441*** (3.819)	0.442*** (3.831)	0.384*** (3.316)	0.376*** (3.225)	0.391*** (3.305)	0.389** (2.404)
EPU	0.817*** (2.755)	0.092 (0.216)	0.235 (0.550)	0.299 (0.695)	0.424 (0.967)	1.014 (1.565)
US EPU		1.432** (2.413)	0.834 (1.385)	0.685 (1.115)	0.650 (1.048)	-0.067 (-0.075)
Δ CLI			-4.794*** (-5.295)	-4.680*** (-5.130)	-5.257*** (-5.512)	-6.094*** (-4.556)
Δ MCI				-0.063 (-1.404)	-0.065 (-1.430)	-0.069 (-1.111)
Oil Volatility					-0.314** (-2.108)	-0.070 (-0.315)
US IP Growth						-0.017*** (-2.856)
Observations	3,600	3,600	3,600	3,600	3,600	3,600
R^2	0.065	0.067	0.075	0.076	0.077	0.081

Note: See notes to Table D.5. EPU and US EPU are economic policy uncertainty indices by Baker et al. (2016), Δ CLI is the change in the composite leading indicator, Δ MCI is the change in Michigan's Consumer Sentiment Index, and oil volatility is from West Texas Intermediate (WTI). t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.7: Dynamic Panel Regressions (Bond Maturity - Seven)

	Dependent Variable: IP Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
US Bond Vol (1-lag)	-0.557** (-2.393)		-0.789*** (-2.948)	-0.776*** (-2.882)	-0.775*** (-2.893)	-0.617** (-2.302)
US Bond Vol (2-lag)	-1.705*** (-7.379)		-0.989*** (-3.545)	-0.975*** (-3.476)	-0.895*** (-3.176)	-0.944*** (-3.370)
US Stock Vol (1-lag)		0.054 (1.254)	0.151*** (3.097)	0.151*** (3.096)	0.205*** (3.298)	0.190*** (3.079)
US Stock Vol (2-lag)		-0.370*** (-8.667)	-0.278*** (-5.625)	-0.274*** (-5.523)	-0.199*** (-3.310)	-0.128** (-2.106)
IP Growth (1-lag)	-0.096*** (-6.032)	-0.091*** (-5.754)	-0.101*** (-6.389)	-0.101*** (-6.398)	-0.104*** (-6.552)	-0.123*** (-7.723)
IP Growth (2-lag)	0.076*** (4.442)	0.083*** (4.776)	0.073*** (4.308)	0.074*** (4.317)	0.072*** (4.243)	0.052*** (3.005)
Bond Vol (1-lag)				-0.014 (-0.341)	0.000 (0.006)	0.014 (0.331)
Bond Vol (2-lag)				-0.025 (-0.595)	-0.009 (-0.213)	0.002 (0.059)
Stock Vol (1-lag)					-0.059 (-1.281)	-0.061 (-1.329)
Stock Vol (2-lag)					-0.090* (-1.935)	-0.060 (-1.297)
Δ Interest Rate (1-lag)						0.441*** (4.119)
Δ Interest Rate (2-lag)						0.392*** (3.683)
Observations	3,600	3,600	3,600	3,600	3,600	3,600
R^2	0.048	0.045	0.056	0.056	0.057	0.068

Note: This table presents the results from a least square dummy variable (LSDV) regression with HAC standard errors (Equation (1)). We select 2 lags for each country based on the SBIC. We consider bonds with maturities of five to seven years (seven). The time period is 1998M3-2017M2. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.8: Dynamic Panel Regressions with Additional Controls (Bond Maturity - Seven)

	Dependent Variable: IP Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
US Bond Vol (1-lag)	-0.627** (-2.180)	-0.721** (-2.499)	-0.634** (-2.207)	-0.658** (-2.275)	-0.584** (-1.965)	-0.610 (-1.483)
US Bond Vol (2-lag)	-0.856*** (-2.834)	-0.828*** (-2.747)	-0.788*** (-2.611)	-0.783*** (-2.584)	-0.721** (-2.326)	-0.805* (-1.876)
US Stock Vol (1-lag)	0.198*** (2.988)	0.173** (2.564)	0.126* (1.859)	0.126* (1.846)	0.117* (1.697)	0.104 (1.080)
US Stock Vol (2-lag)	-0.109* (-1.664)	-0.117* (-1.787)	-0.127* (-1.947)	-0.121* (-1.836)	-0.124* (-1.859)	-0.138 (-1.496)
IP Growth (1-lag)	-0.122*** (-7.687)	-0.120*** (-7.557)	-0.132*** (-8.328)	-0.134*** (-8.457)	-0.137*** (-8.632)	-0.139*** (-8.709)
IP Growth (2-lag)	0.052*** (2.832)	0.054*** (2.923)	0.039** (2.071)	0.041** (2.183)	0.039** (2.029)	0.039 (1.493)
Bond Vol (1-lag)	0.015 (0.333)	0.014 (0.306)	0.020 (0.449)	0.019 (0.416)	0.017 (0.370)	0.012 (0.191)
Bond Vol (2-lag)	0.005 (0.110)	0.002 (0.048)	0.011 (0.239)	0.011 (0.257)	0.009 (0.202)	0.006 (0.099)
Stock Vol (1-lag)	-0.088* (-1.757)	-0.078 (-1.556)	-0.093* (-1.846)	-0.093* (-1.834)	-0.087* (-1.695)	-0.100 (-1.412)
Stock Vol (2-lag)	-0.071 (-1.422)	-0.076 (-1.516)	-0.072 (-1.454)	-0.071 (-1.425)	-0.069 (-1.360)	-0.069 (-0.980)
Δ Interest Rate (1-lag)	0.463*** (4.032)	0.480*** (4.177)	0.485*** (4.222)	0.482*** (4.170)	0.495*** (4.255)	0.462*** (2.854)
Δ Interest Rate (2-lag)	0.418*** (3.662)	0.419*** (3.678)	0.373*** (3.257)	0.365*** (3.168)	0.374*** (3.208)	0.372** (2.305)
EPU	0.888*** (3.153)	0.166 (0.407)	0.327 (0.802)	0.400 (0.969)	0.456 (1.090)	1.089* (1.736)
US EPU		1.462** (2.516)	0.878 (1.483)	0.725 (1.197)	0.703 (1.153)	-0.044 (-0.049)
Δ CLI			-4.388*** (-4.893)	-4.269*** (-4.719)	-4.655*** (-4.902)	-5.478*** (-4.075)
Δ MCI				-0.062 (-1.404)	-0.064 (-1.426)	-0.068 (-1.092)
Oil Volatility					-0.194 (-1.296)	0.063 (0.279)
US IP Growth						-0.017*** (-2.962)
Observations	3,600	3,600	3,600	3,600	3,600	3,600
R^2	0.071	0.073	0.079	0.080	0.080	0.085

Note: See notes to Table D.7. EPU and US EPU are economic policy uncertainty indices by Baker et al. (2016), Δ CLI is the change in the composite leading indicator, Δ MCI is the change in Michigan's Consumer Sentiment Index, and oil volatility is from West Texas Intermediate (WTI). t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.9: Dynamic Panel Regressions (Bond Maturity - Ten)

	Dependent Variable: IP Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
US Bond Vol (1-lag)	-0.635*** (-3.400)		-0.840*** (-3.911)	-0.804*** (-3.710)	-0.793*** (-3.667)	-0.629*** (-2.902)
US Bond Vol (2-lag)	-1.444*** (-7.792)		-0.991*** (-4.450)	-0.998*** (-4.390)	-0.940*** (-4.093)	-0.933*** (-4.074)
US Stock Vol (1-lag)		0.054 (1.254)	0.154*** (3.218)	0.159*** (3.315)	0.192*** (3.101)	0.175*** (2.847)
US Stock Vol (2-lag)		-0.370*** (-8.667)	-0.223*** (-4.481)	-0.221*** (-4.415)	-0.160*** (-2.650)	-0.101* (-1.648)
IP Growth (1-lag)	-0.107*** (-6.761)	-0.091*** (-5.754)	-0.111*** (-6.964)	-0.110*** (-6.910)	-0.111*** (-7.008)	-0.127*** (-7.996)
IP Growth (2-lag)	0.064*** (3.796)	0.083*** (4.776)	0.064*** (3.755)	0.063*** (3.726)	0.063*** (3.711)	0.045*** (2.645)
Bond Vol (1-lag)				-0.076 (-1.081)	-0.060 (-0.839)	-0.048 (-0.674)
Bond Vol (2-lag)				0.027 (0.379)	0.056 (0.771)	0.066 (0.911)
Stock Vol (1-lag)					-0.035 (-0.751)	-0.039 (-0.834)
Stock Vol (2-lag)					-0.081* (-1.744)	-0.056 (-1.203)
Δ Interest Rate (1-lag)						0.428*** (4.018)
Δ Interest Rate (2-lag)						0.337*** (3.164)
Observations	3,600	3,600	3,600	3,600	3,600	3,600
R^2	0.057	0.045	0.063	0.063	0.063	0.073

Note: This table presents the results from a least square dummy variable (LSDV) regression with HAC standard errors (Equation (1)). We select 2 lags for each country based on the SBIC. We consider bonds with maturities of seven to ten years (ten). The time period is 1998M3-2017M2. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.10: Dynamic Panel Regressions with Additional Controls (Bond Maturity - Ten)

	Dependent Variable: IP Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
US Bond Vol (1-lag)	-0.677*** (-2.925)	-0.710*** (-3.076)	-0.610*** (-2.650)	-0.618*** (-2.672)	-0.577** (-2.435)	-0.563* (-1.713)
US Bond Vol (2-lag)	-0.892*** (-3.647)	-0.860*** (-3.520)	-0.783*** (-3.192)	-0.791*** (-3.202)	-0.742*** (-2.924)	-0.820** (-2.333)
US Stock Vol (1-lag)	0.188*** (2.872)	0.163** (2.424)	0.119* (1.766)	0.118* (1.733)	0.113 (1.637)	0.095 (0.983)
US Stock Vol (2-lag)	-0.074 (-1.136)	-0.081 (-1.243)	-0.097 (-1.470)	-0.089 (-1.339)	-0.092 (-1.377)	-0.105 (-1.129)
IP Growth (1-lag)	-0.127*** (-7.980)	-0.125*** (-7.858)	-0.135*** (-8.480)	-0.137*** (-8.623)	-0.139*** (-8.760)	-0.141*** (-8.859)
IP Growth (2-lag)	0.045** (2.493)	0.047*** (2.585)	0.035* (1.871)	0.037** (1.990)	0.035* (1.867)	0.036 (1.370)
Bond Vol (1-lag)	-0.050 (-0.653)	-0.060 (-0.788)	-0.064 (-0.841)	-0.063 (-0.821)	-0.068 (-0.877)	-0.074 (-0.685)
Bond Vol (2-lag)	0.071 (0.916)	0.060 (0.771)	0.061 (0.777)	0.062 (0.783)	0.055 (0.692)	0.053 (0.482)
Stock Vol (1-lag)	-0.067 (-1.339)	-0.059 (-1.163)	-0.073 (-1.439)	-0.073 (-1.433)	-0.068 (-1.311)	-0.081 (-1.133)
Stock Vol (2-lag)	-0.068 (-1.373)	-0.071 (-1.432)	-0.069 (-1.386)	-0.068 (-1.350)	-0.066 (-1.301)	-0.065 (-0.931)
Δ Interest Rate (1-lag)	0.453*** (3.986)	0.468*** (4.108)	0.473*** (4.151)	0.470*** (4.095)	0.481*** (4.173)	0.449*** (2.796)
Δ Interest Rate (2-lag)	0.364*** (3.202)	0.367*** (3.225)	0.331*** (2.898)	0.322*** (2.806)	0.332*** (2.847)	0.329** (2.042)
EPU	1.016*** (3.661)	0.384 (0.958)	0.508 (1.261)	0.588 (1.440)	0.616 (1.494)	1.266** (2.046)
US EPU		1.285** (2.230)	0.783 (1.332)	0.616 (1.022)	0.619 (1.019)	-0.143 (-0.163)
Δ CLI			-3.918*** (-4.392)	-3.791*** (-4.205)	-4.138*** (-4.348)	-4.966*** (-3.681)
Δ MCI				-0.065 (-1.475)	-0.066 (-1.491)	-0.070 (-1.139)
Oil Volatility					-0.162 (-1.089)	0.090 (0.403)
US IP Growth						-0.017*** (-2.980)
Observations	3,600	3,600	3,600	3,600	3,600	3,600
R^2	0.076	0.078	0.083	0.083	0.084	0.089

Note: See notes to Table D.9. EPU and US EPU are economic policy uncertainty indices by Baker et al. (2016), Δ CLI is the change in the composite leading indicator, Δ MCI is the change in Michigan's Consumer Sentiment Index, and oil volatility is from West Texas Intermediate (WTI). t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Alternative Panel VAR Specifications

For additional robustness, we investigate how international industrial production growth responds to sudden one-standard-deviation increases in the innovations of U.S. and international financial market volatility at different horizons in an alternative panel VAR framework. In the spirit of Love and Zicchino (2006), we estimate the following panel VAR system using GMM,

$$Y_{i,t} = \alpha + \sum_{j=1}^2 \beta_j Y_{i,t-j} + f_i + \varepsilon_{i,t} \quad (3)$$

where $Y_{i,t}$ denotes a four-variable vector (panel) of (1) international industrial production growth, (2) change in the international interest rate, (3) U.S. bond market volatility, and (4) one of the following: U.S. stock market volatility, international stock market volatility, or international bond market volatility. We control for country fixed effects (f_i). Because the fixed effects are usually correlated with the regressors as noted by Love and Zicchino (2006), we use a forward mean-differencing procedure and use lagged regressors as instruments for our GMM estimation.²³ We also use the Cholesky decomposition and generate 95% confidence bands from 1,000 Monte Carlo simulations.²⁴

Figure E.1 presents our panel VAR results for three separate four-variable VARs. The top sub-figure includes U.S. stock market volatility (US Stock) as the fourth variable in the VAR, the middle sub-figure includes international bond market volatility (Intl Bond) as the fourth variable in the VAR, and the bottom sub-figure includes international stock market volatility (Intl Stock) as the fourth variable in the VAR. We consider bonds with maturities of one or more years (one plus). The impulse response functions suggest that a one-standard-deviation increase in U.S. bond market volatility appears to have a negative spillover effect on international industrial production growth over time and this effect is significant even after controlling for the other U.S. and international financial market variables. Moreover, the response of industrial production growth to a U.S. bond market volatility shock is more negative than the responses of industrial production growth to the other financial market volatility shocks.

Next, we consider the corresponding impulse response analysis from Section 4 with different bond maturities using our sets of four-variable panel VARs. Consistent with the dynamic panel regression results, the volatility of U.S. bonds with longer maturities have a stronger impact on international industrial production growth than volatility of U.S. bonds with shorter maturities (Figures E.2 and E.3).

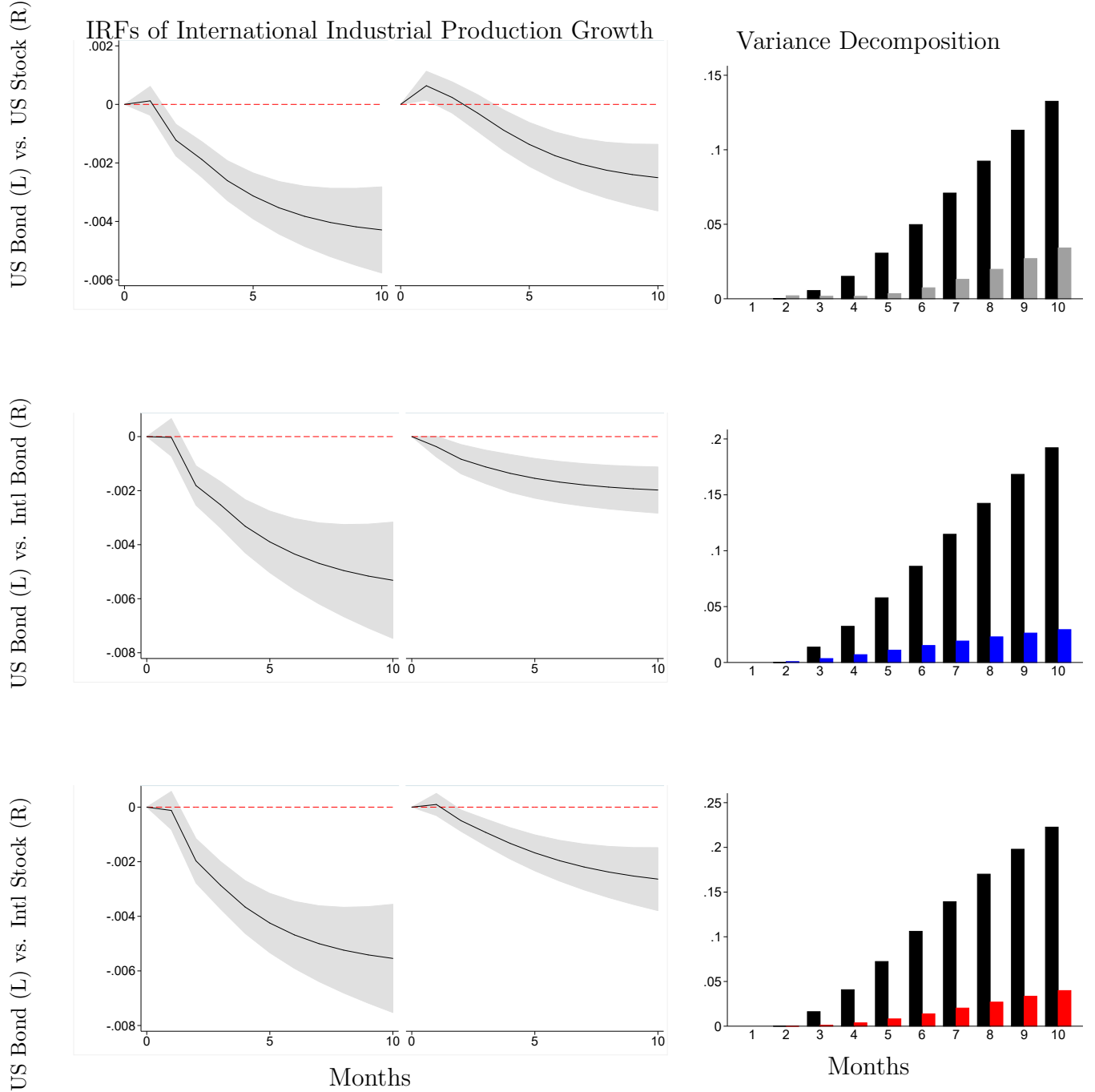
We also decompose the variance contribution of each financial market volatility variable

²³We use up to 6 lags as instruments for our GMM estimation, but our generated impulse response functions are not sensitive to a larger set of instruments.

²⁴Our results are robust to 4,000 Monte Carlo simulations.

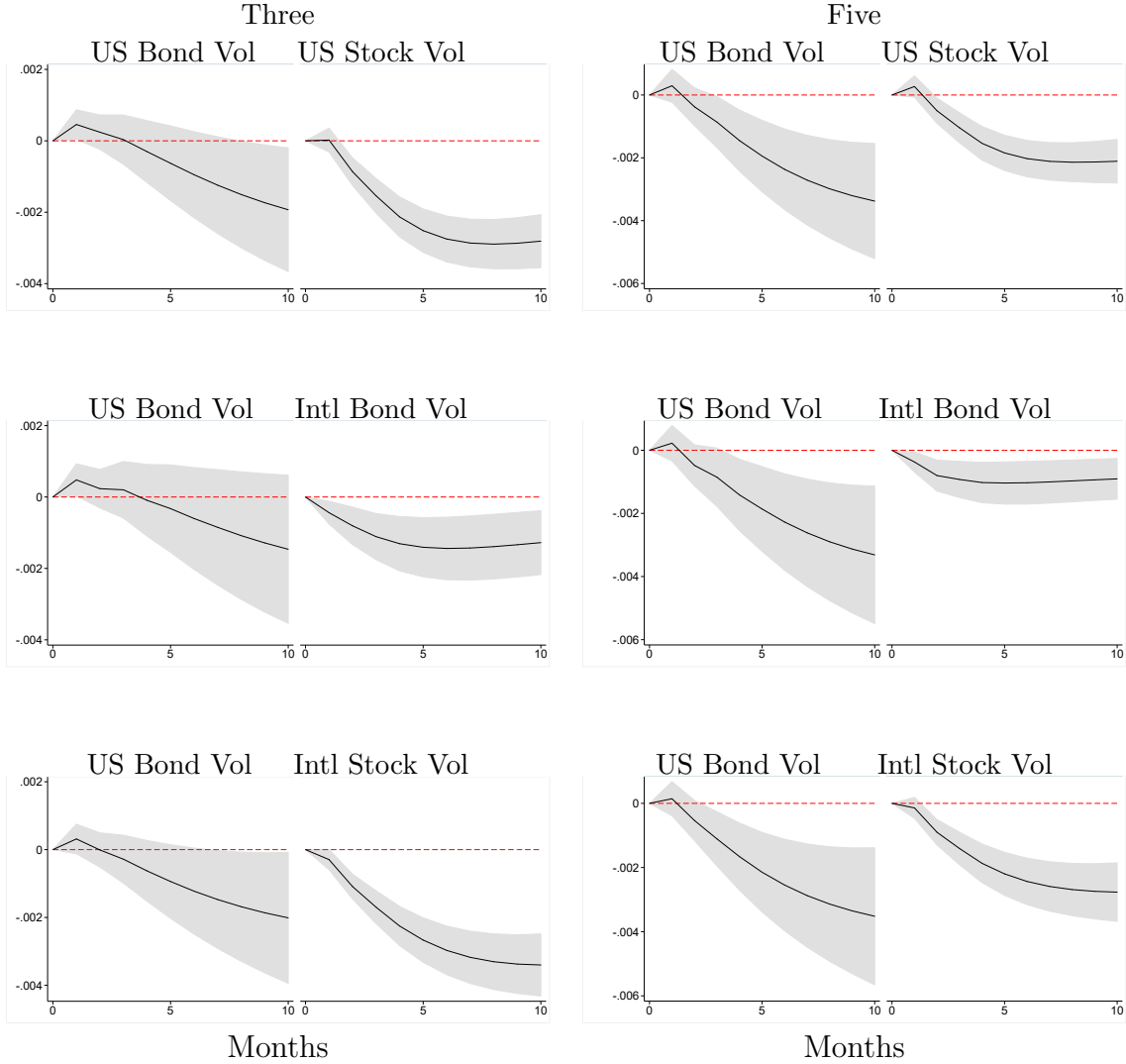
(U.S./international bond and stock markets) to international industrial production growth using the same four-variable panel VAR setup, but with different maturities for bonds (three, five, seven, and ten years). Figure E.4 presents the contributions of these four volatility measures from one to ten months after the shocks. Again, we use black for U.S. bond market volatility, grey for U.S. stock market volatility, blue for international bond market volatility, and red for international stock market volatility. Consistent with our baseline panel VAR results, volatility of U.S. bonds with longer maturities is able to explain more variance of cross-country international industrial production growth than volatility of U.S. bonds with shorter horizons.

Figure E.1: IRFs and Variance Decomposition of International Industrial Production Growth to Financial Market Volatility Shocks (Bond Maturity - One Plus)



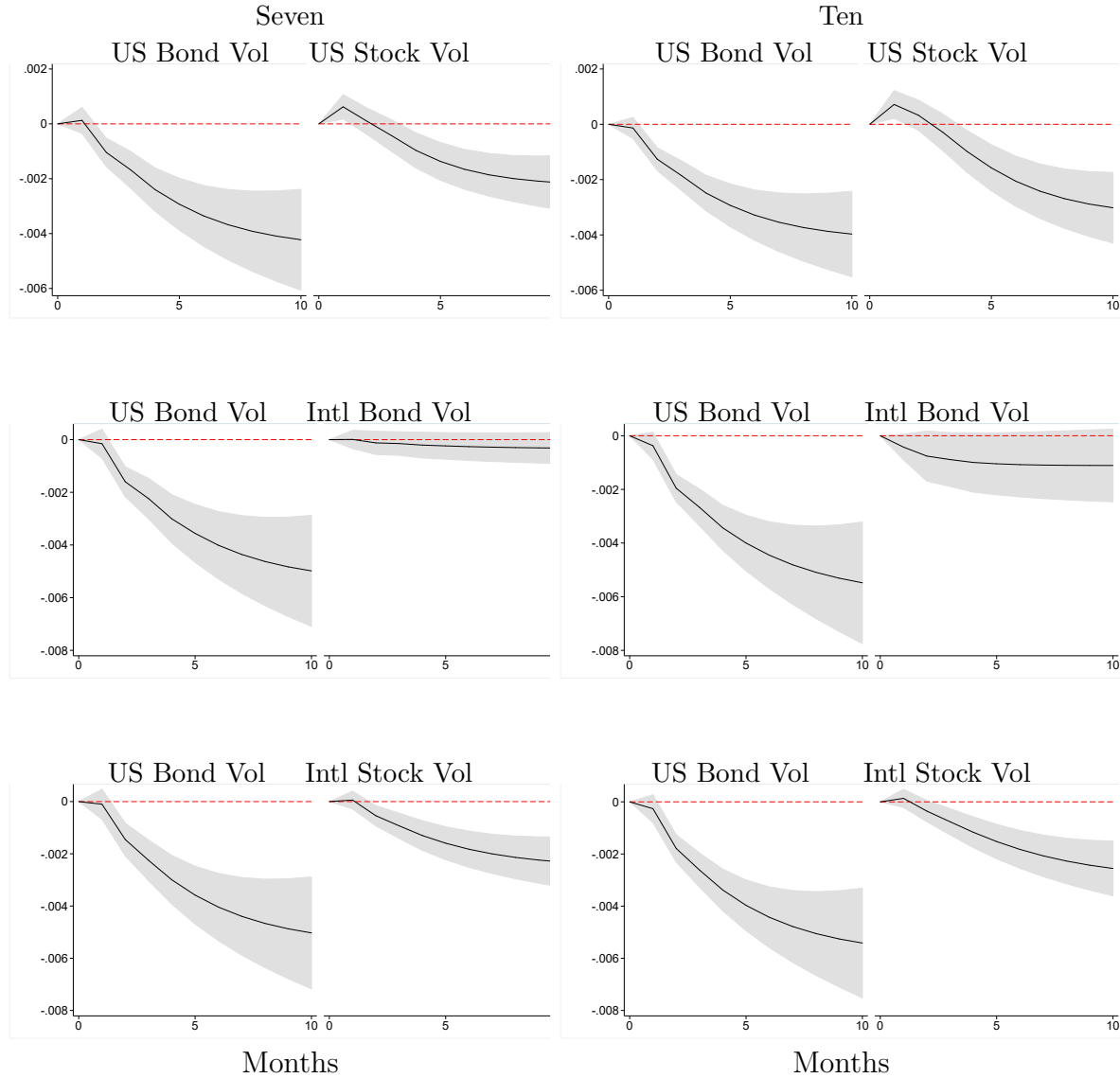
Note: This figure plots the impulse response functions of international industrial production growth to one-standard-deviation increases in the innovations of the U.S. and international financial market variables for up to ten months after the shocks. These are generated from a four-variable panel VAR where the first three variables are always international industrial production growth, international interest rate change, and U.S. bond volatility, and the last variable is denoted in the left vertical axis label (U.S. stock volatility, international bond volatility, or international stock volatility). The first IRF column is the response of international industrial production growth to a U.S. bond volatility shock and the second IRF column is the response of international industrial production growth to one of the other financial volatility shocks. 95% confidence bands from 1,000 Monte Carlo simulations are included in grey. The associated variance decomposition is plotted in the last column. We use black for U.S. bond volatility, grey for U.S. stock volatility, blue for international bond volatility, and red for international stock volatility. We consider bonds with maturities of one or more years (one plus). The time period is 1998M3-2017M2.

Figure E.2: IRFs of International Industrial Production Growth to Financial Market Volatility Shocks (Bond Maturities - Three, Five)



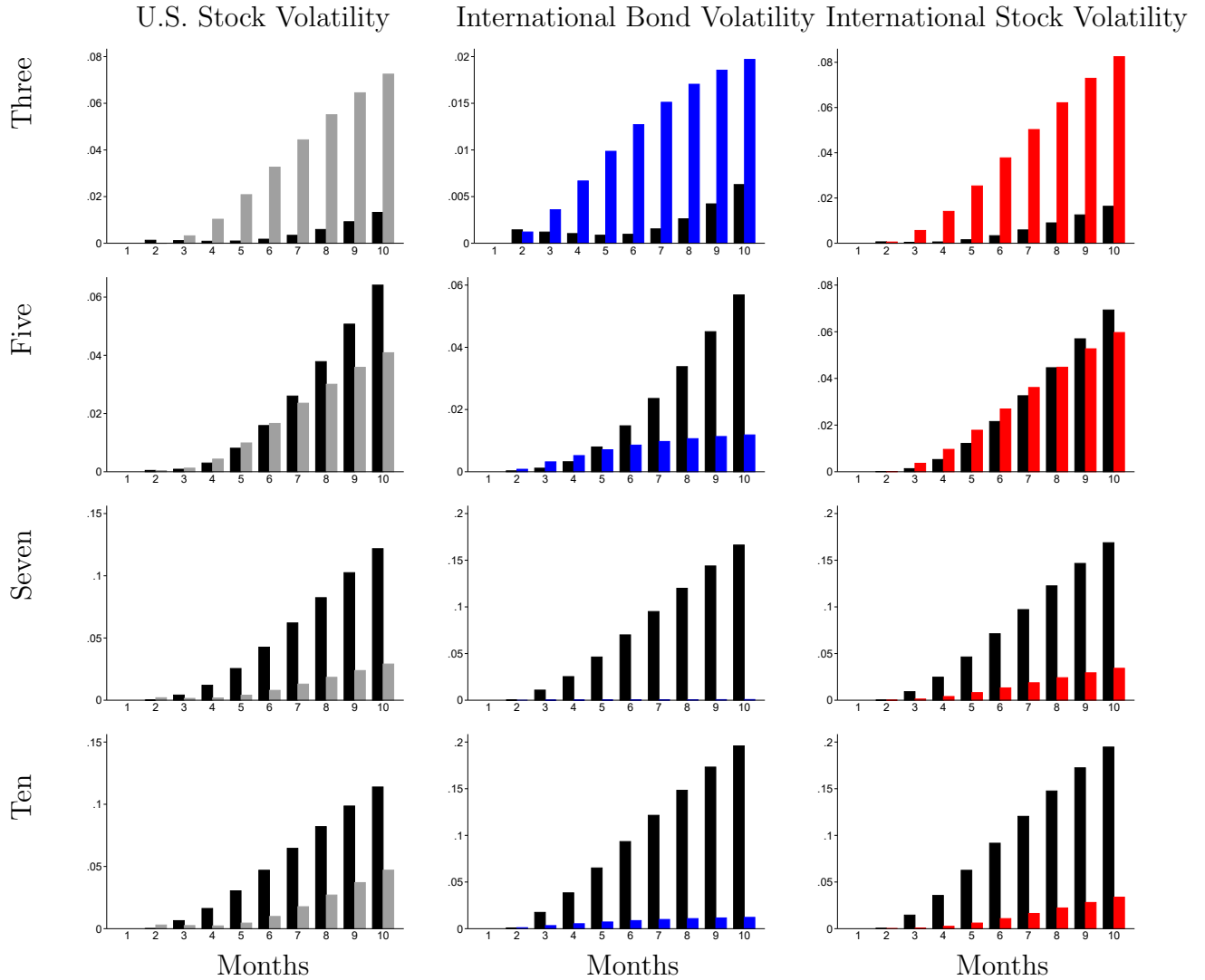
Note: This figure plots the impulse response functions of international industrial production growth to one-standard-deviation increases in the innovations of the U.S. and international financial market variables for up to ten months after the shocks. These are generated from a four-variable panel VAR where the first three variables are always international industrial production growth, international interest rate change, and U.S. bond market volatility, and the last variable is U.S. stock market volatility, international bond market volatility, or international stock market volatility. We consider bonds with maturities of one to three years (three) and three to five years (five). 95% confidence bands from 1,000 Monte Carlo simulations are included in grey. The time period is 1998M3-2017M2.

Figure E.3: IRFs of International Industrial Production Growth to Financial Market Volatility Shocks (Bond Maturities - Seven, Ten)



Note: This figure plots the impulse response functions of international industrial production growth to one-standard-deviation increases in the innovations of the U.S. and international financial market variables for up to ten months after the shocks. These are generated from a four-variable panel VAR where the first three variables are always international industrial production growth, international interest rate change, and U.S. bond market volatility, and the last variable is U.S. stock market volatility, international bond market volatility, or international stock market volatility. We consider bonds with maturities of five to seven years (seven) and seven to ten years (ten). 95% confidence bands from 1,000 Monte Carlo simulations are included in grey. The time period is 1998M3-2017M2.

Figure E.4: Variance Decomposition of International Industrial Production Growth (Bond Maturities - Three, Five, Seven, Ten)



Note: This figure plots the variance decomposition of international industrial production growth for up to ten months after the shocks, which is generated from a four-variable panel VAR where the first three variables are always international industrial production growth, international interest rate change, and U.S. bond market volatility, and the last variable is U.S. stock market volatility, international bond market volatility, or international stock market volatility. We use black for U.S. bond market volatility, grey for U.S. stock market volatility, blue for international bond market volatility, and red for international stock market volatility. We consider bonds with maturities of one to three years (three), three to five years (five), five to seven years (seven), and seven to ten years (ten). The time period is 1998M3-2017M2.